

CSCI 598H

Applied Computer Vision for Engineering Systems

Lecture 1-2: Course Overview

Dr. Minghan Li

Assistant Professor

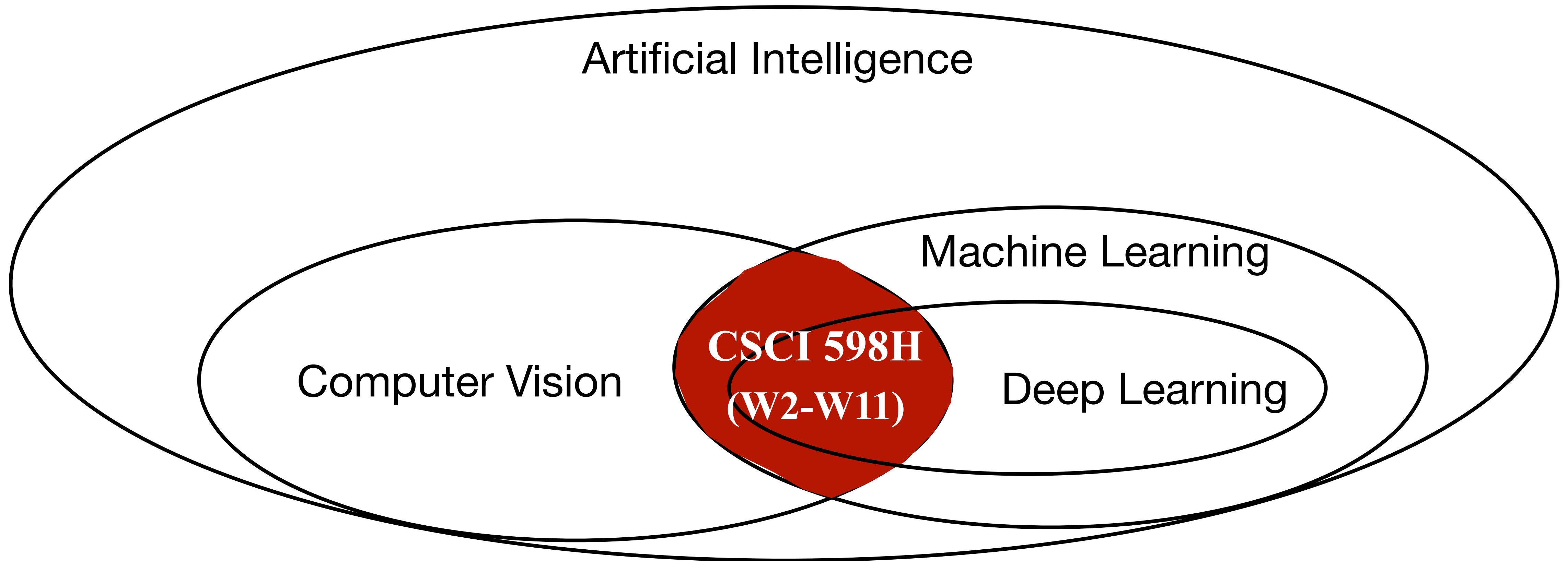
Computer Science Department

Colorado School of Mines

Aug/27/2026

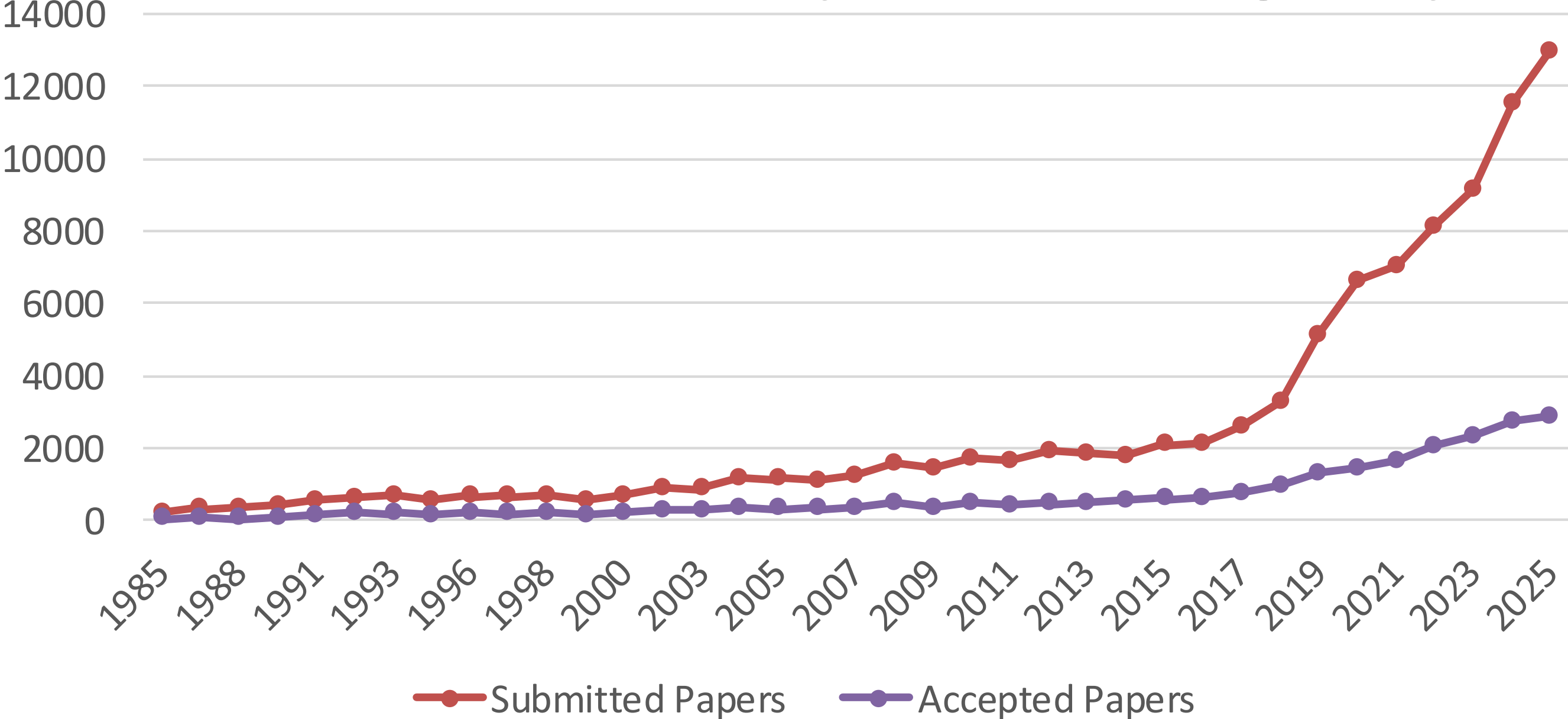


Recap: Deep Learning for Computer Vision

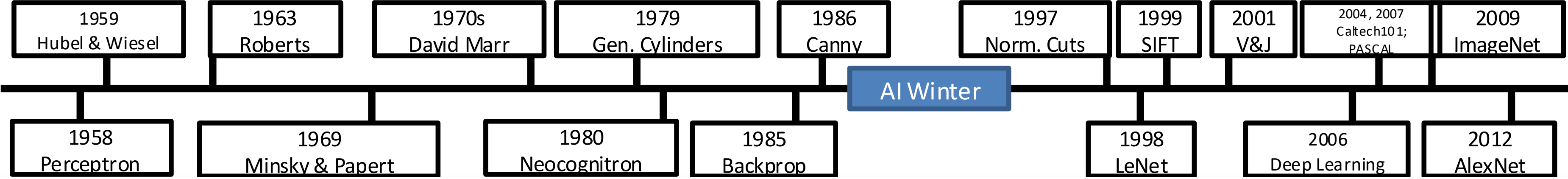


Recap: A brief history of computer vision

2012 to Present: Deep Learning Explosion



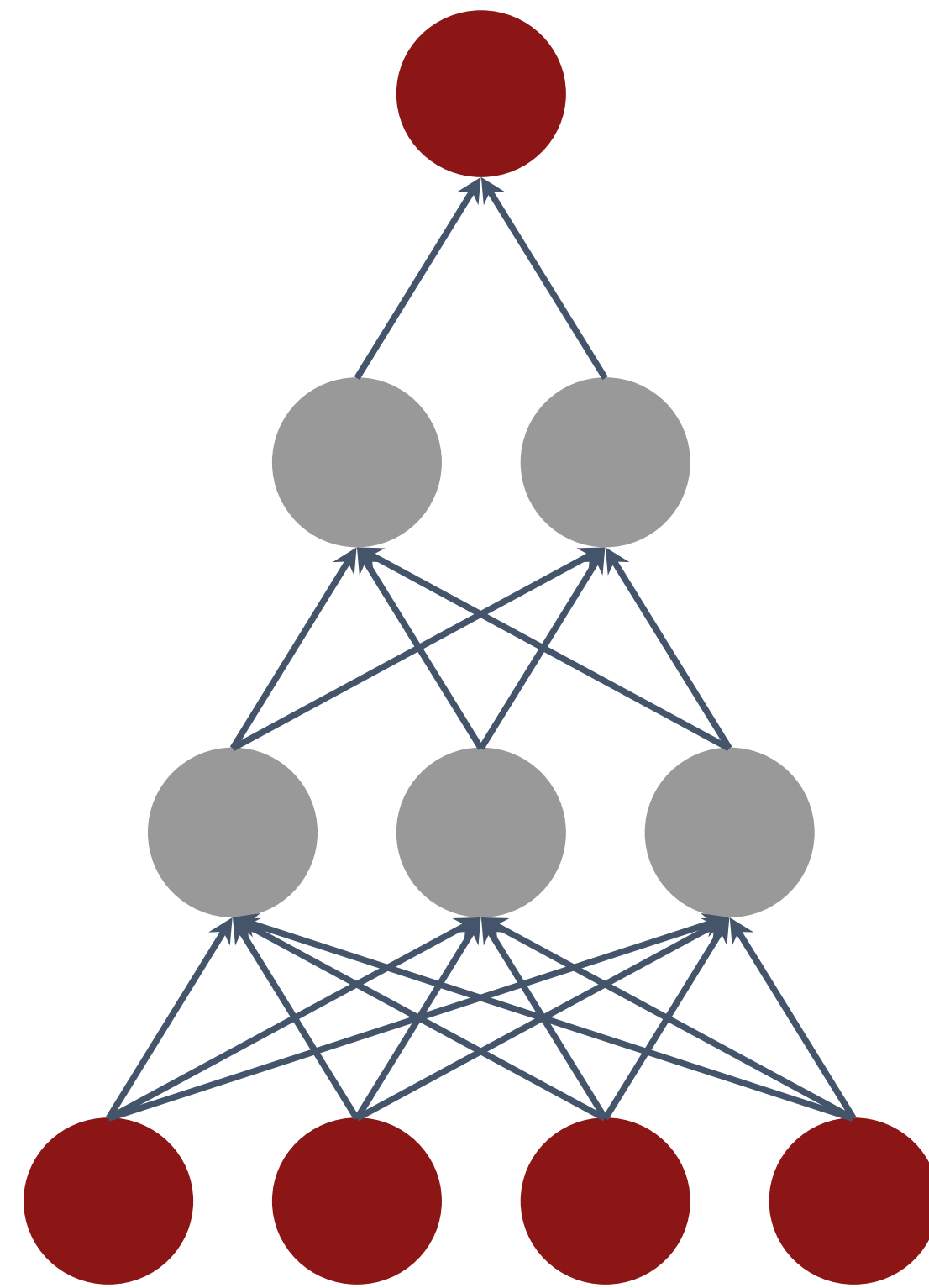
Publications at top Computer Vision conference



Recap: A brief history of computer vision



Data



Algorithms



Computation

Today's agenda

- A brief history of computer vision (CV)
- CSCI 598H course overview
- CSCI 598H course administration

CSCI 598H Course Overview

Weeks 2-4

Weeks 5-11

Weeks 12-16

Module 1

Deep Learning Basics

- Linear Classifier and Optimization
- CNNs and BackPropagation
- RNNs, Attention, Transformers

Module 2

Modern Visual Intelligence

- Perceiving and understanding the visual world
- Generative models and foundation models
- Geometry, 3D vision and spatial understanding

Module 3

Engineering Systems

- Robotics
- Earth science
- Medical imaging
- Energy

CSCI 598H Course Overview



Weeks 2-4

Weeks 5-10

Weeks 11-16

Module 1

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Modern Visual Intelligence

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Module 3

Engineering Systems

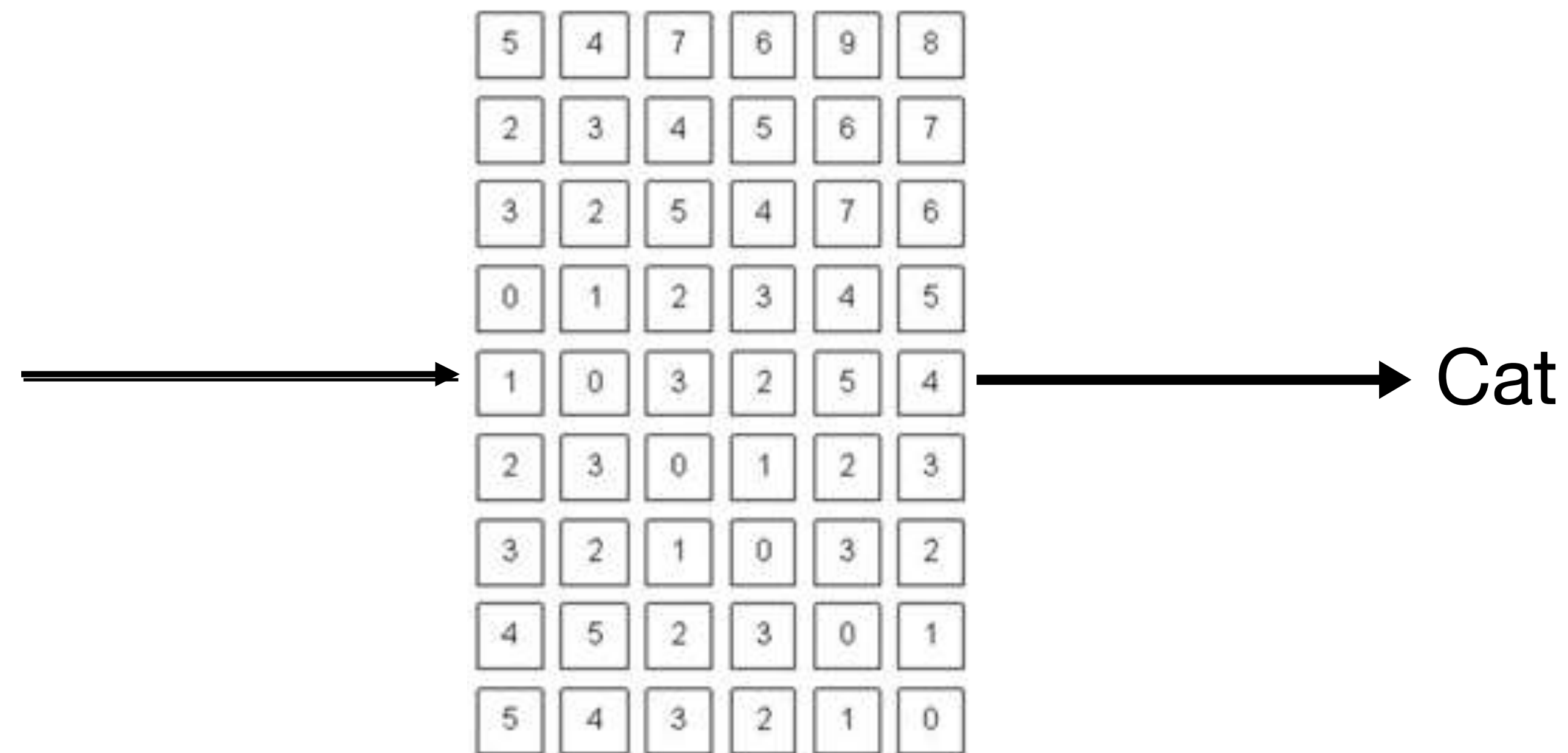
- Robotics
- Earth science
- Medical imaging
- Energy

Deep Learning Basics

- Image Classification: A core task in Computer Vision



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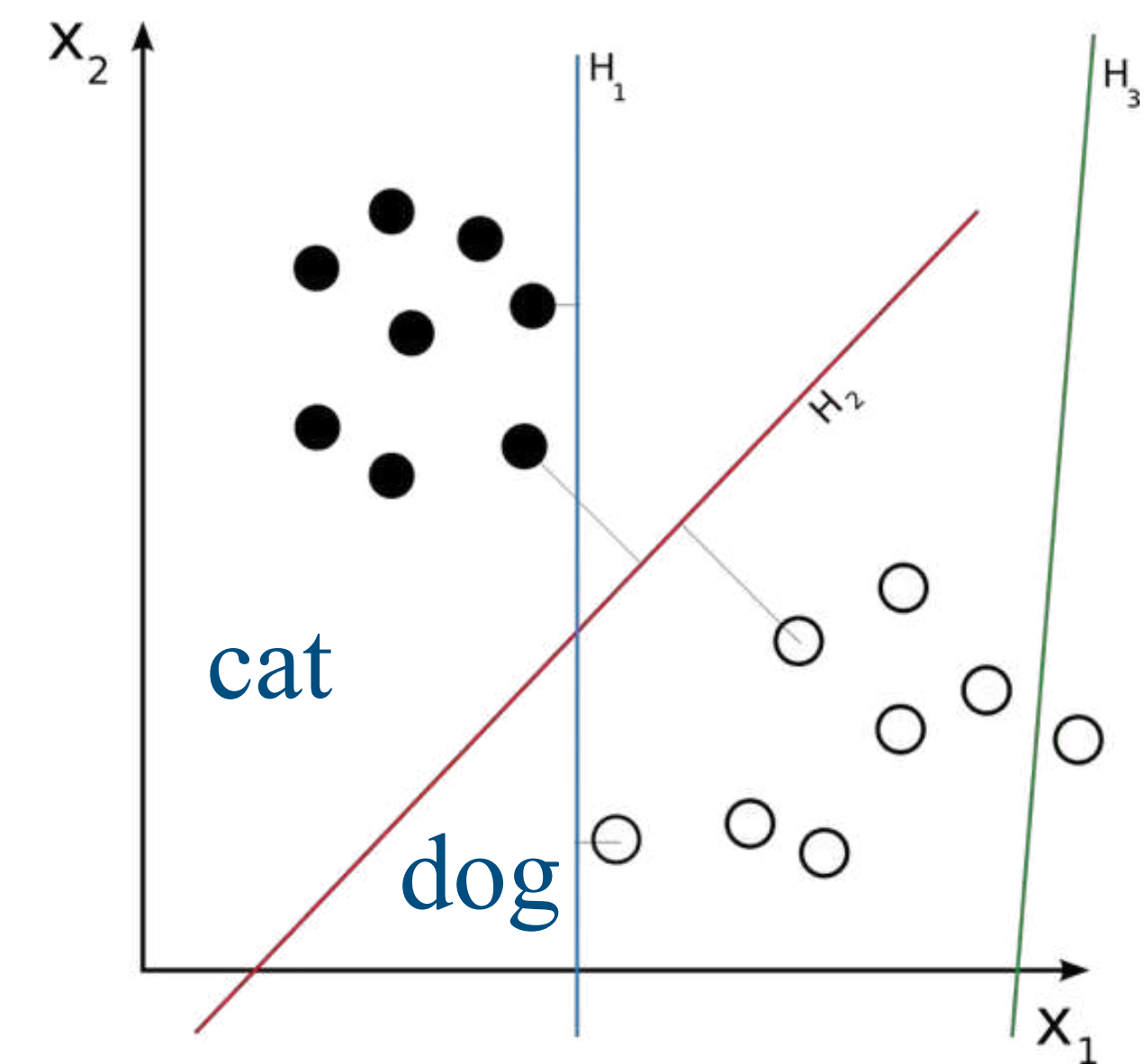
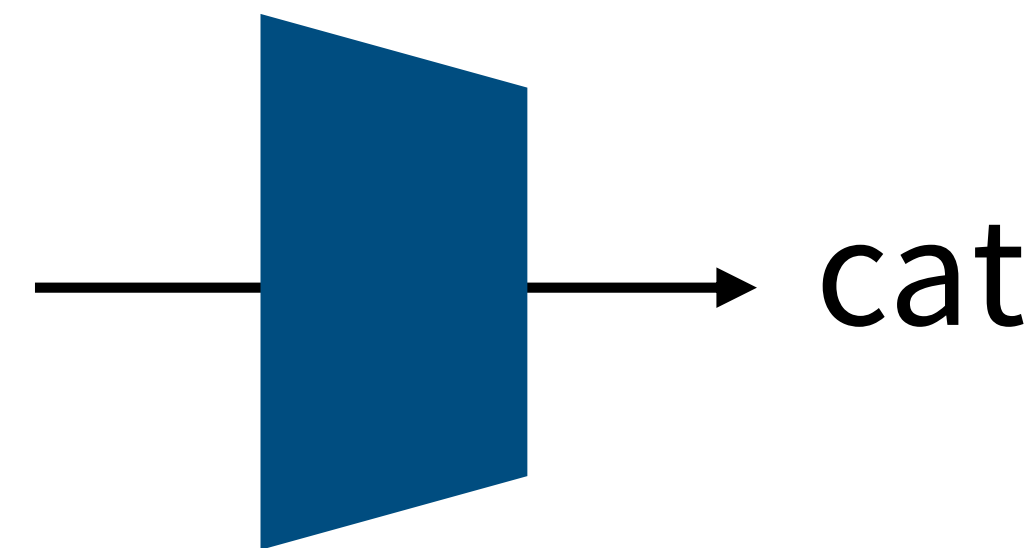
Cat

Deep Learning Basics

- Image Classification: A core task in Computer Vision



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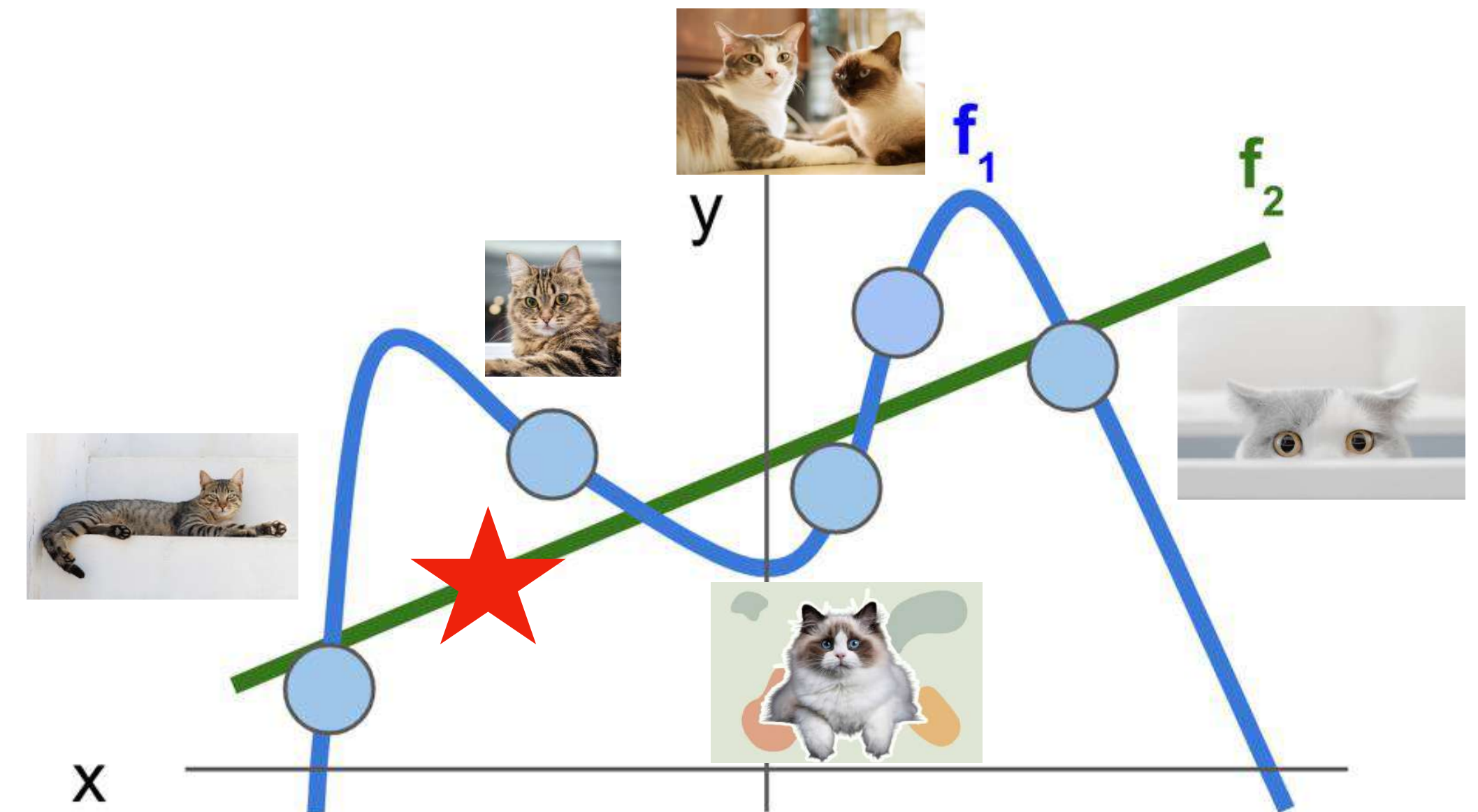
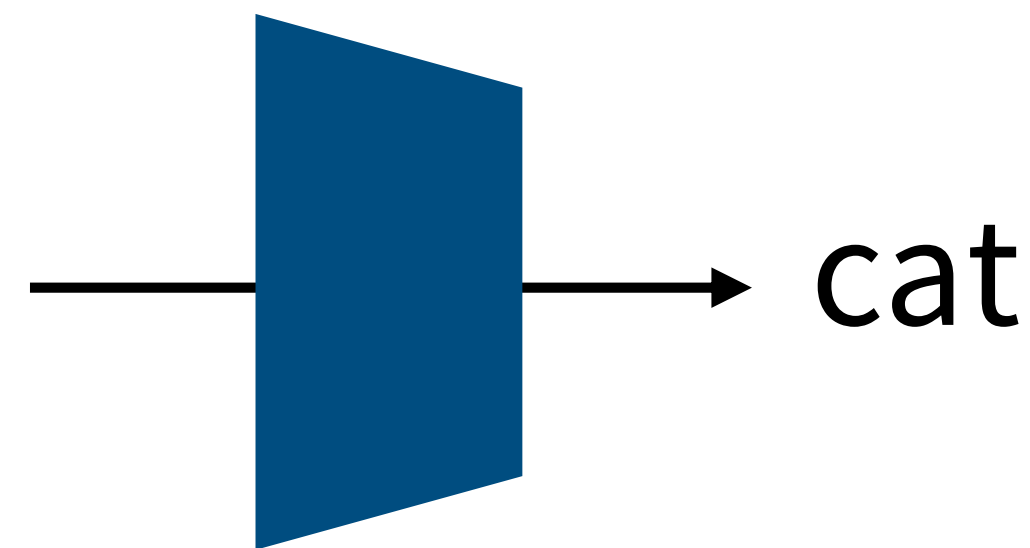
Linear Classifier

Deep Learning Basics

- Image Classification: A core task in Computer Vision



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Regularization & Optimization

A model is useful only if it **learns patterns that generalize beyond the training examples.**

Regularization controls complexity, while optimization searches for good parameters.

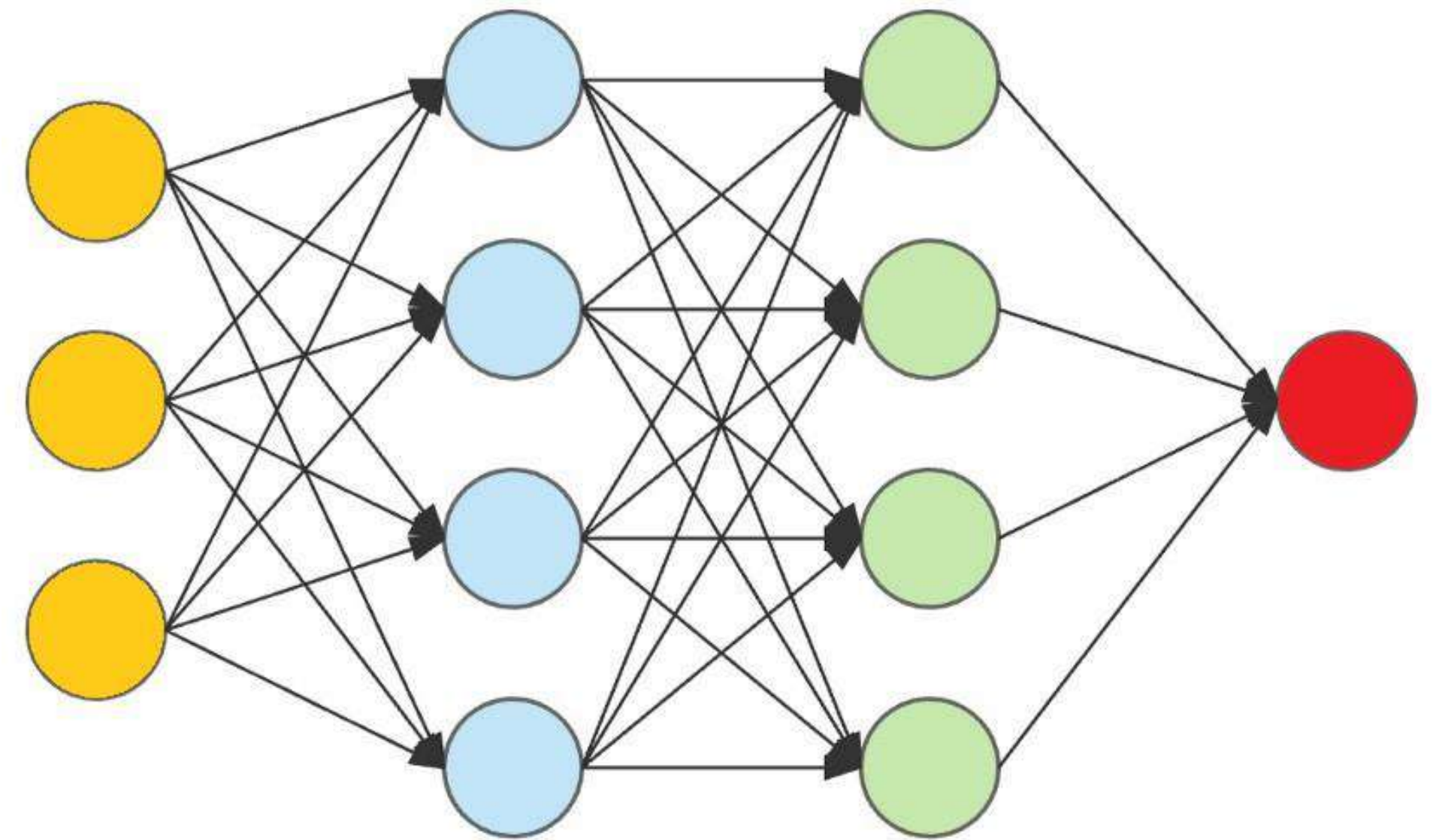
Deep Learning Basics

- Image Classification: A core task in Computer Vision



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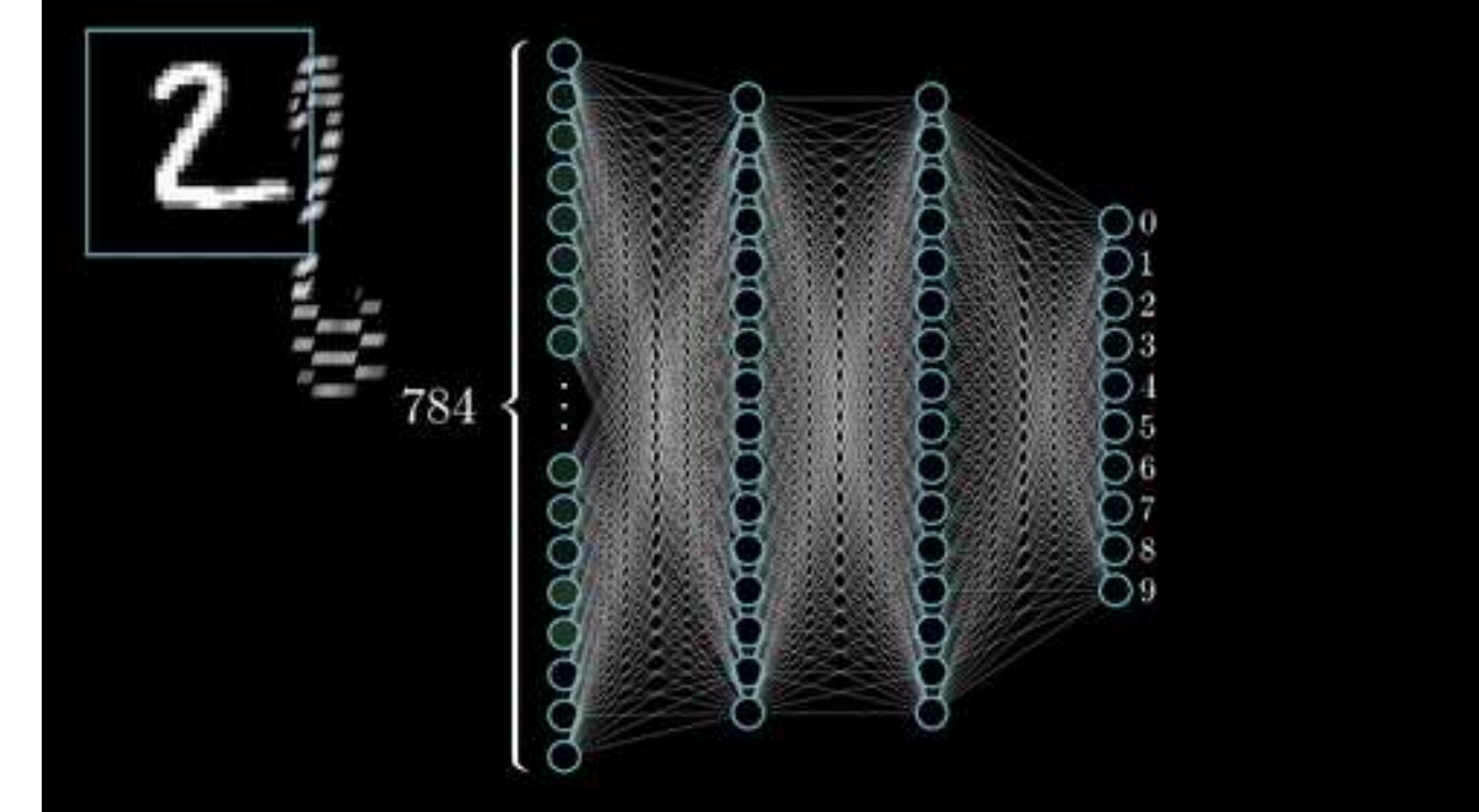
→ cat



Neural Networks

multiple transformations with
nonlinear activation functions

Deep Learning Basics



https://medium.com/@Suraj_Yadav/in-depth-knowledge-of-convolutional-neural-networks-b4bfff8145ab

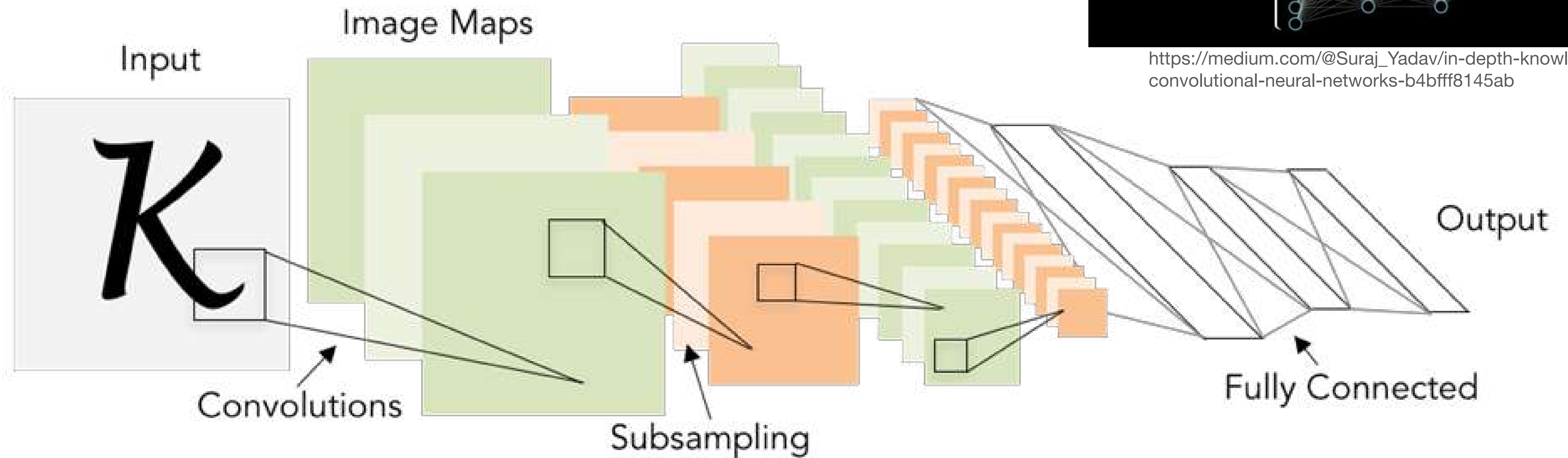
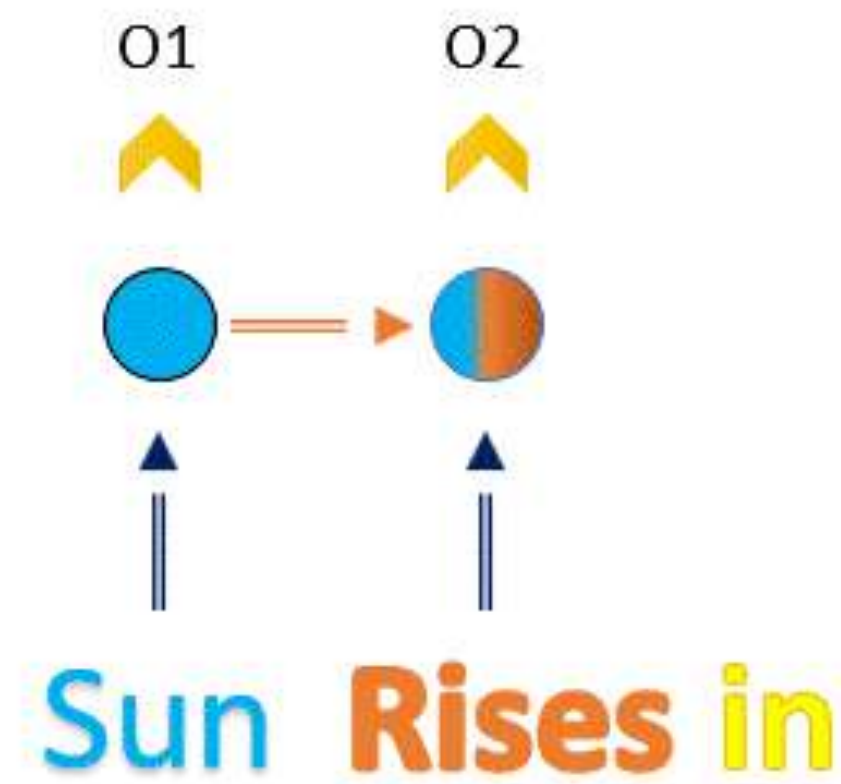


Illustration of LeCun et al. 1998 from CS231n 2017 Lecture 1

Convolutional neural network

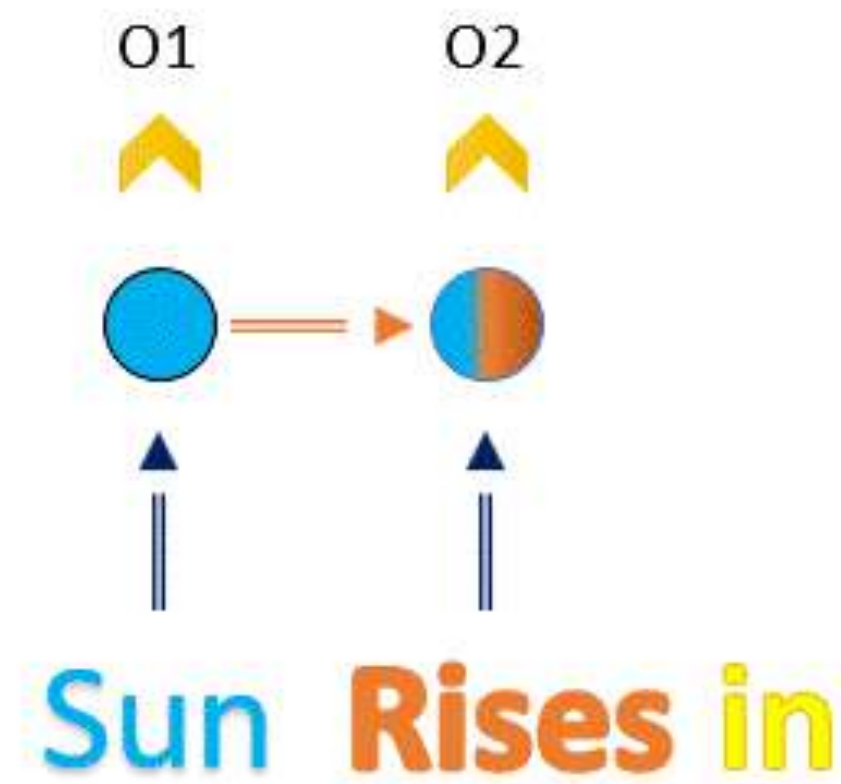
Apply shared filters across spatial locations $\Rightarrow$ decouple repeated features and locations

Deep Learning Basics



Sequence modeling
for time series data
(It evolves over time.)

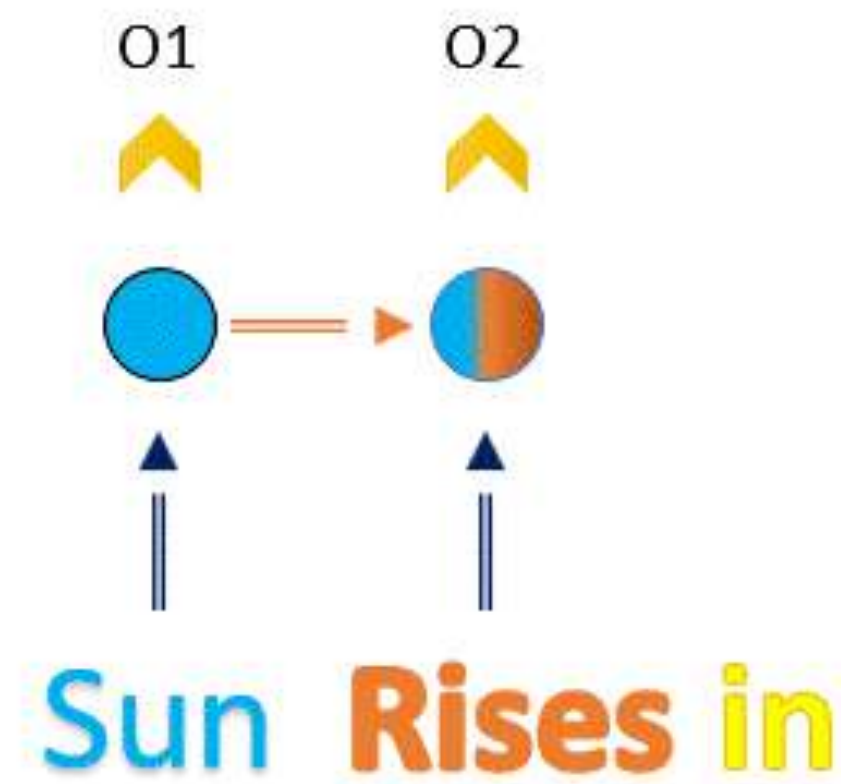
Deep Learning Basics



Sequence modeling
for time series data
(It evolves over time.)

	$t = 1$	$t = 2$	$\dots$	$t = T$	
Language	Sun	Rises	in	the	East

Deep Learning Basics

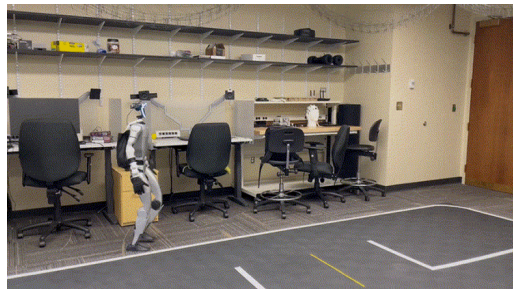
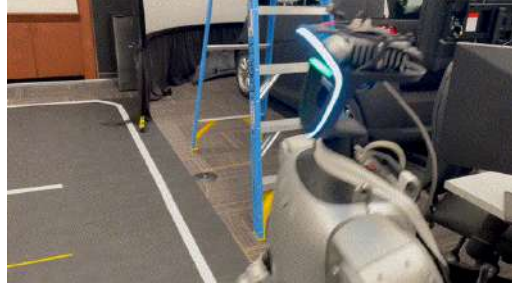


Sequence modeling
for time series data
(It evolves over time.)

Language

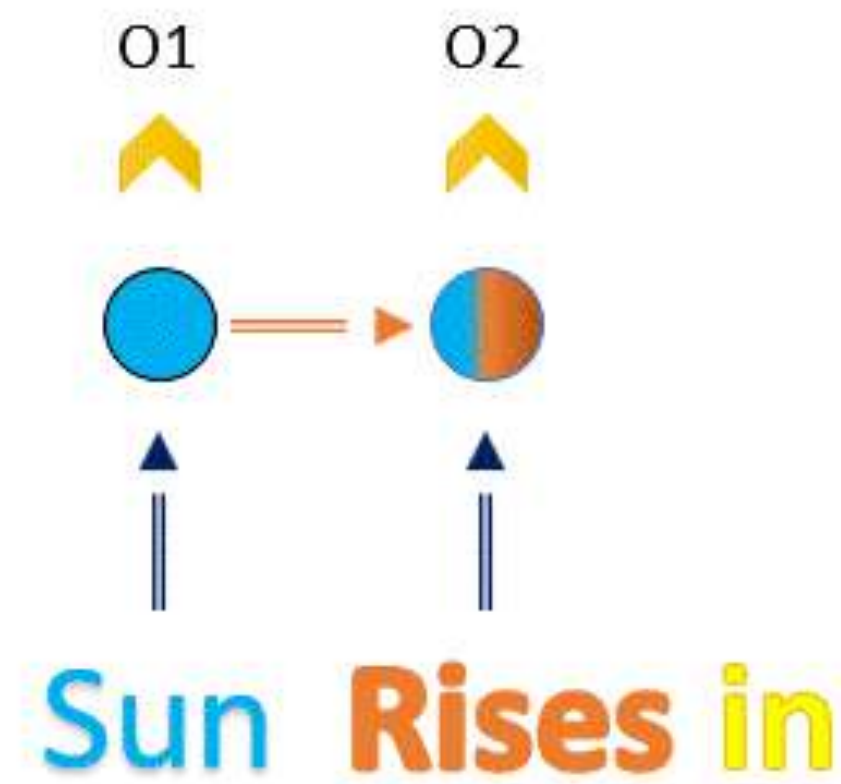
$t = 1$	$t = 2$	$\dots$	$t = T$
Sun	Rises	in	the East

Video

$t = 1$	$t = 2$	$\dots$	$t = T$
			

Others? What else is sequential?

Deep Learning Basics

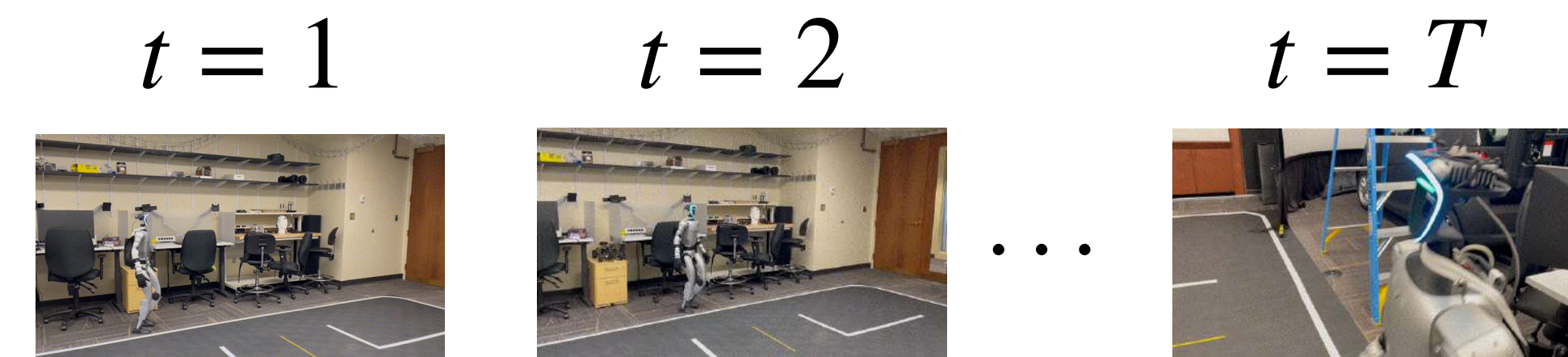


Sequence modeling
for time series data
(It evolves over time.)

Language

$t = 1$ $t = 2$ $\dots$ $t = T$
Sun Rises in the East

Video

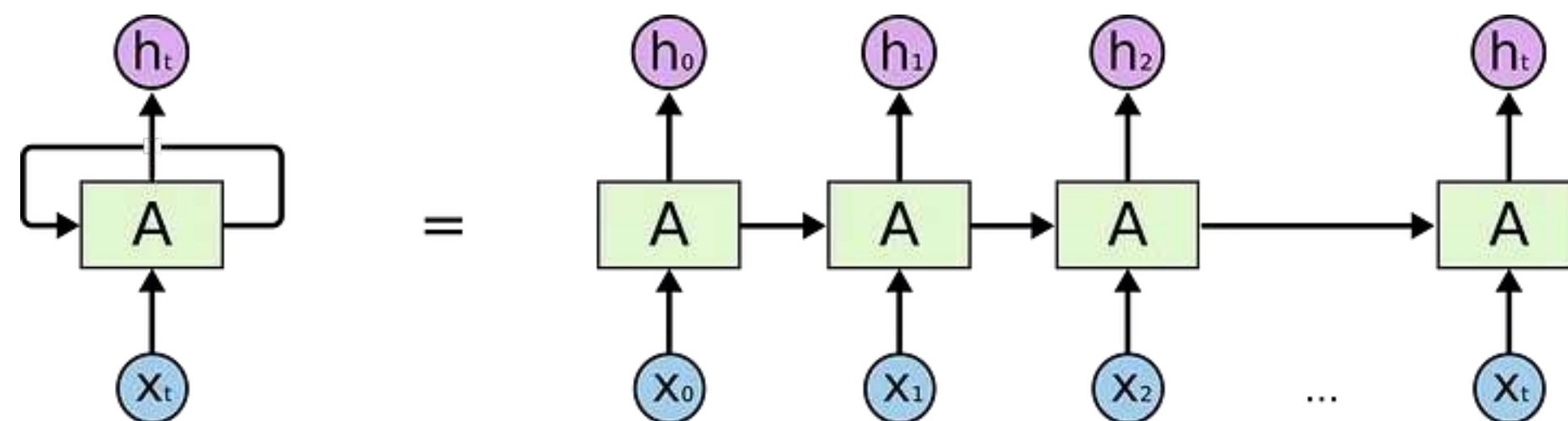


Others?

- Weather forecasting
- Stock trading
- More cases in Engineering

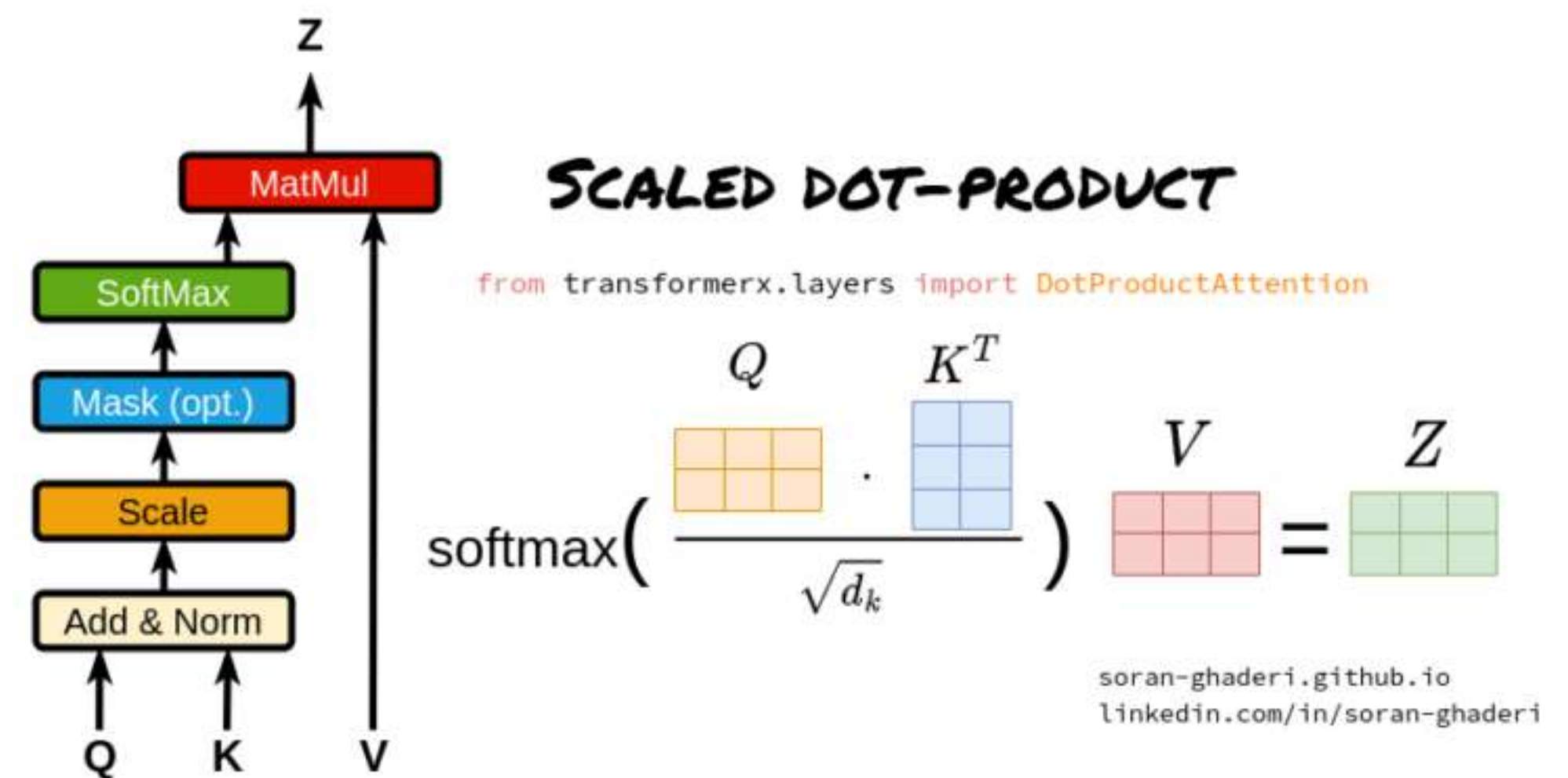
Deep Learning Basics

Long sequences $\Rightarrow$ optimization and memory challenges



Recurrent neural network

long-range dependencies difficult and training slow



Attention mechanism / Transformer

directly compare tokens
& enable parallel training

CSCI 598H Course Overview



Weeks 2-4

Weeks 5-11

Weeks 12-16

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Module 2

Modern Visual Intelligence

- Perceiving and understanding the visual world
- Foundation models and generative models
- Geometry, 3D vision and spatial understanding

Module 3

Engineering Systems

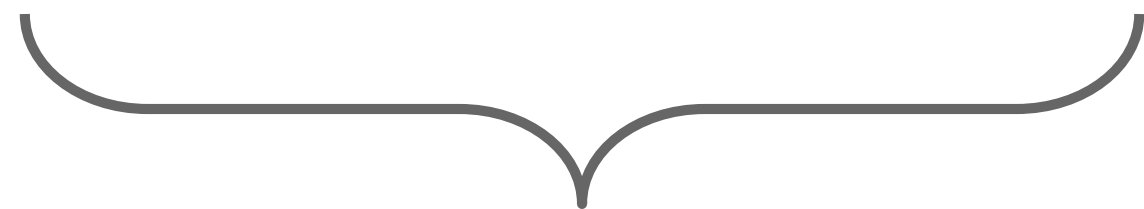
- Robotics
- Earth science
- Medical imaging
- Energy

Perceiving and understanding the visual world

Classification



CAT



No spatial extent

Entire image → Label

Perceiving and understanding the visual world

Classification

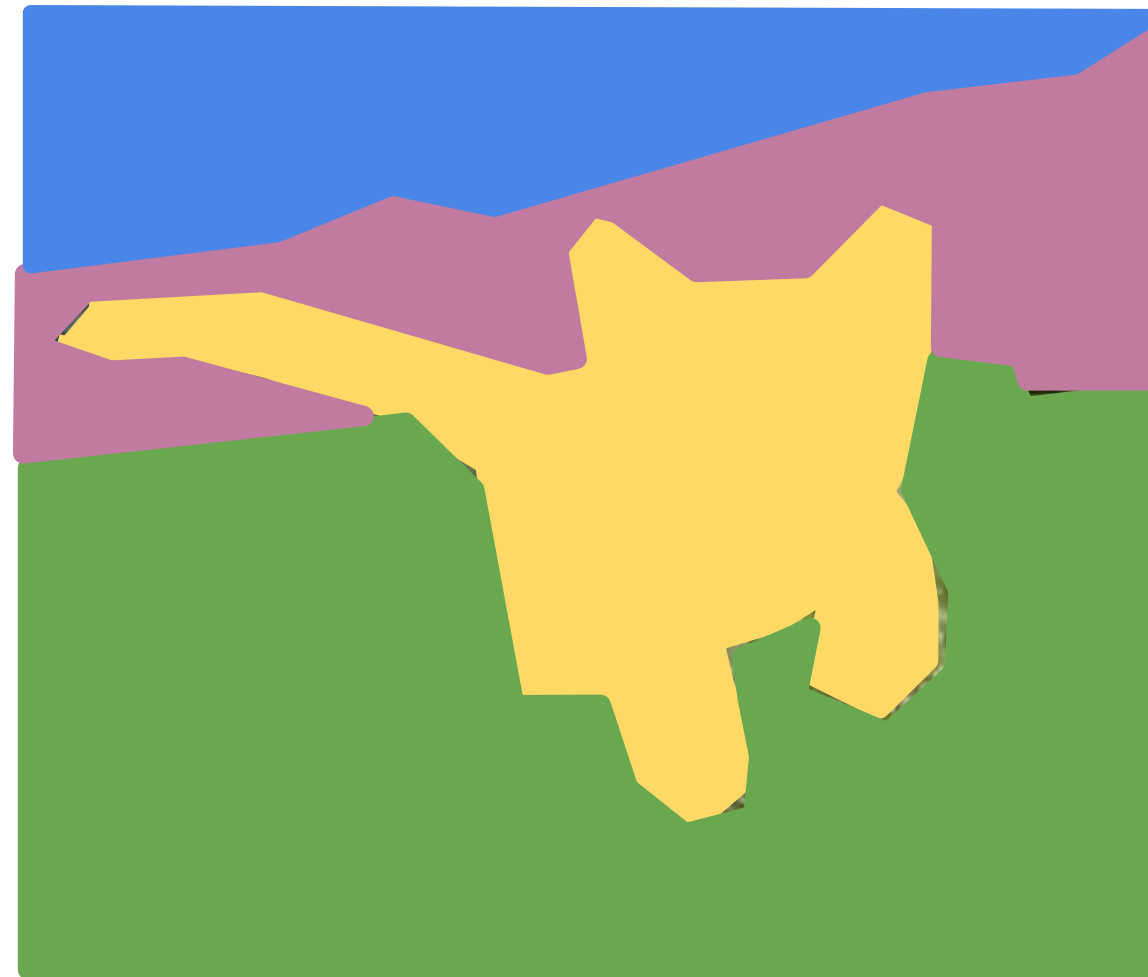


CAT

No spatial extent

Entire image → Label

Semantic Segmentation



GRASS, CAT, TREE,
SKY

No objects, just pixels

Each pixel → Label

Perceiving and understanding the visual world

Classification

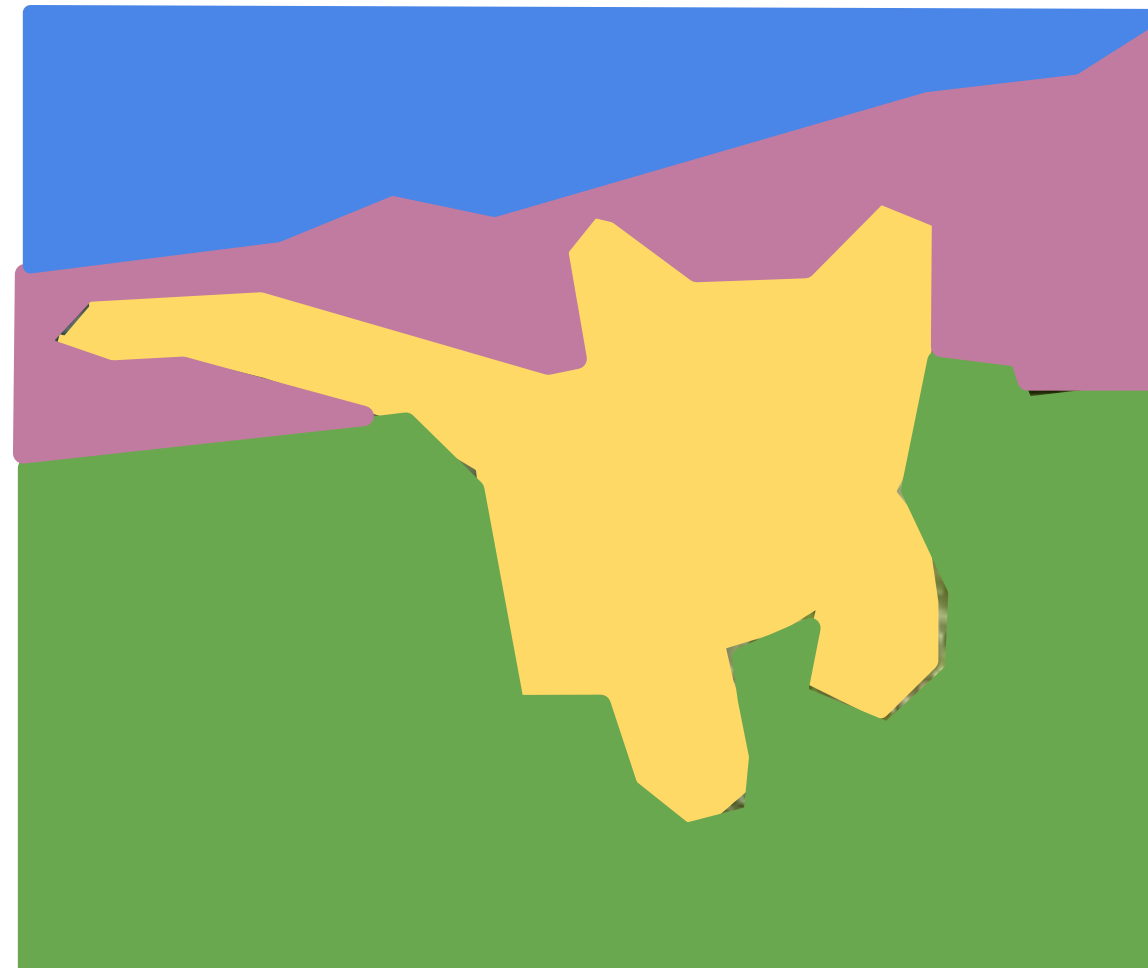


CAT

No spatial extent

Entire image → Label

Semantic Segmentation



GRASS, CAT, TREE,
SKY

No objects, just pixels

Each pixel → Label

Object Detection



DOG, DOG, CAT

Multiple Object

Distinguish separate object instances

Instance Segmentation



DOG, DOG, CAT

[This image is CC0 public domain](#)

Perceiving and understanding the visual world

Video Classification



Swimming? Eating?
(Action classification)

Motion Estimation & Segmentation



Optical Flow



Modified with Segment Anything 2 Demo

Multimodal Video Understanding



Q: Describe this image please

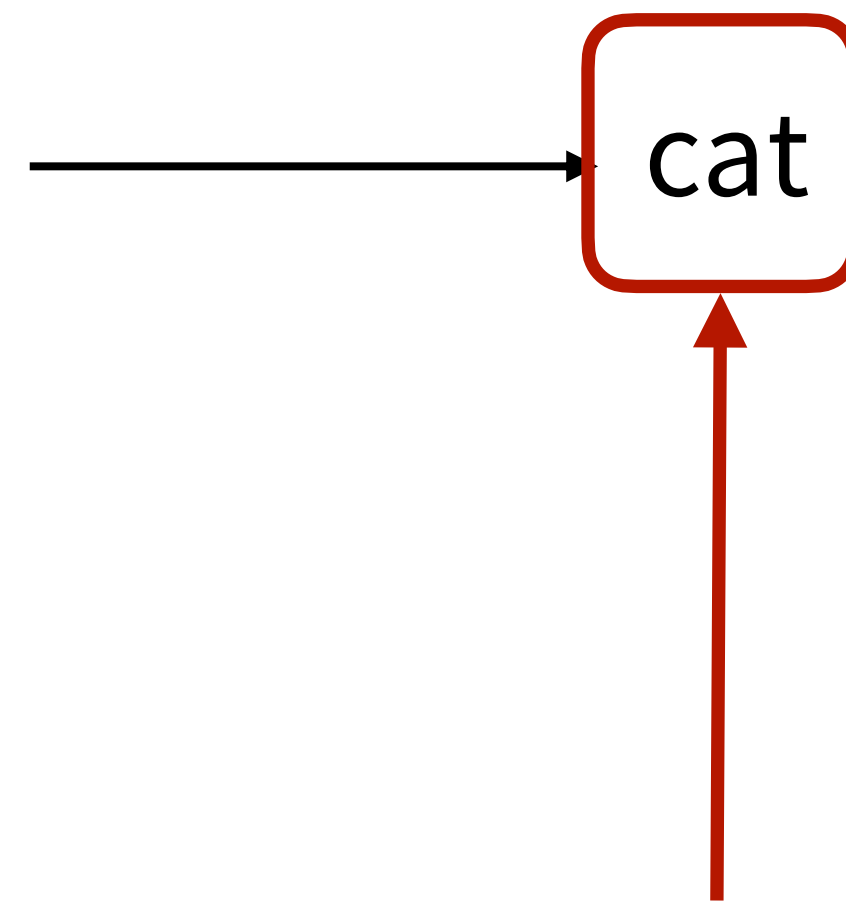
A: A black swan is floating on the water. The background is ...

Foundation models

from supervised learning to *****



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**Label from human
(Supervision)**



Foundation models

from supervised learning to self-supervised learning



This image by Nikita is
licensed under CC-BY 2.0

cat

Label from human
(Supervision)



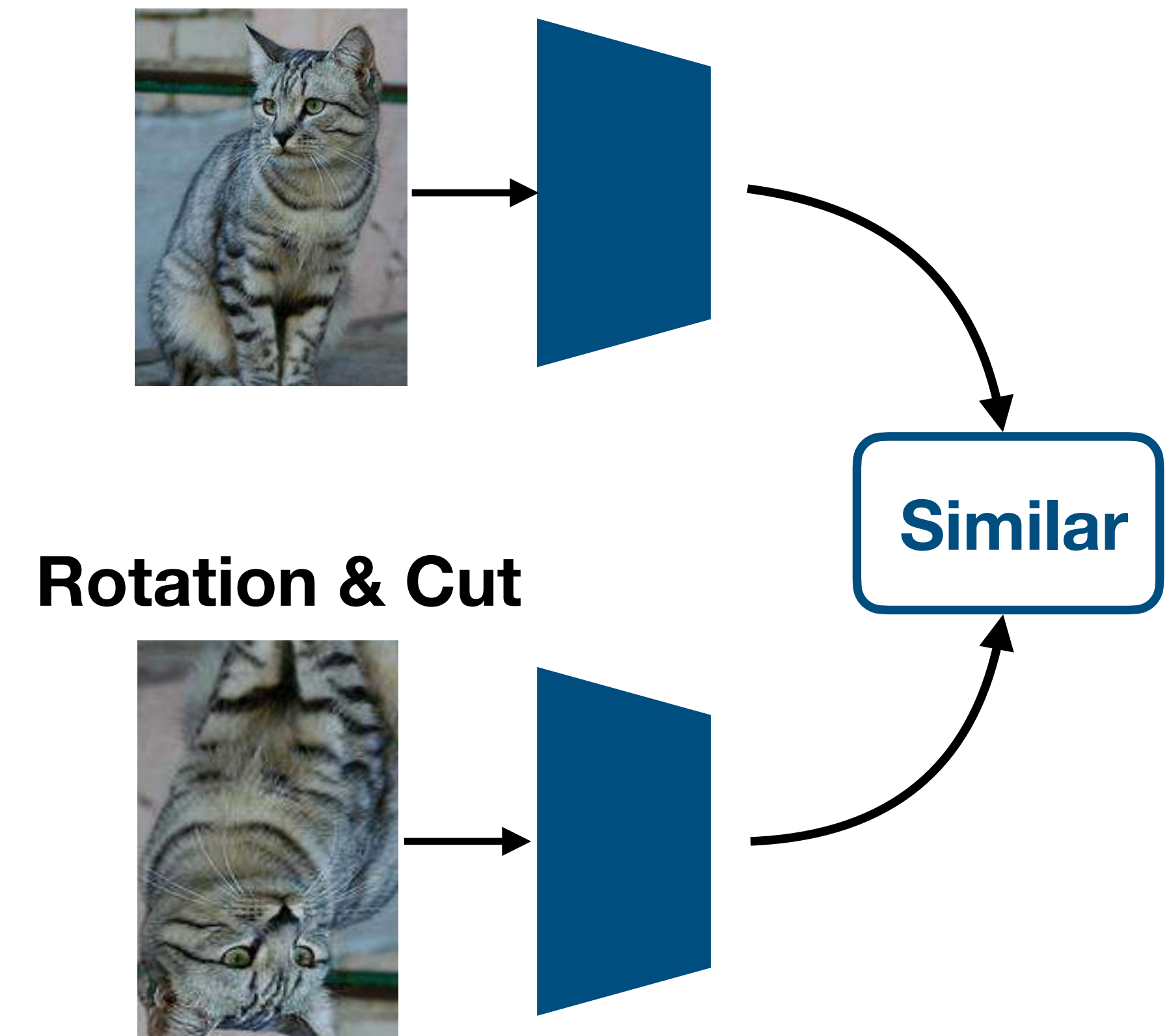
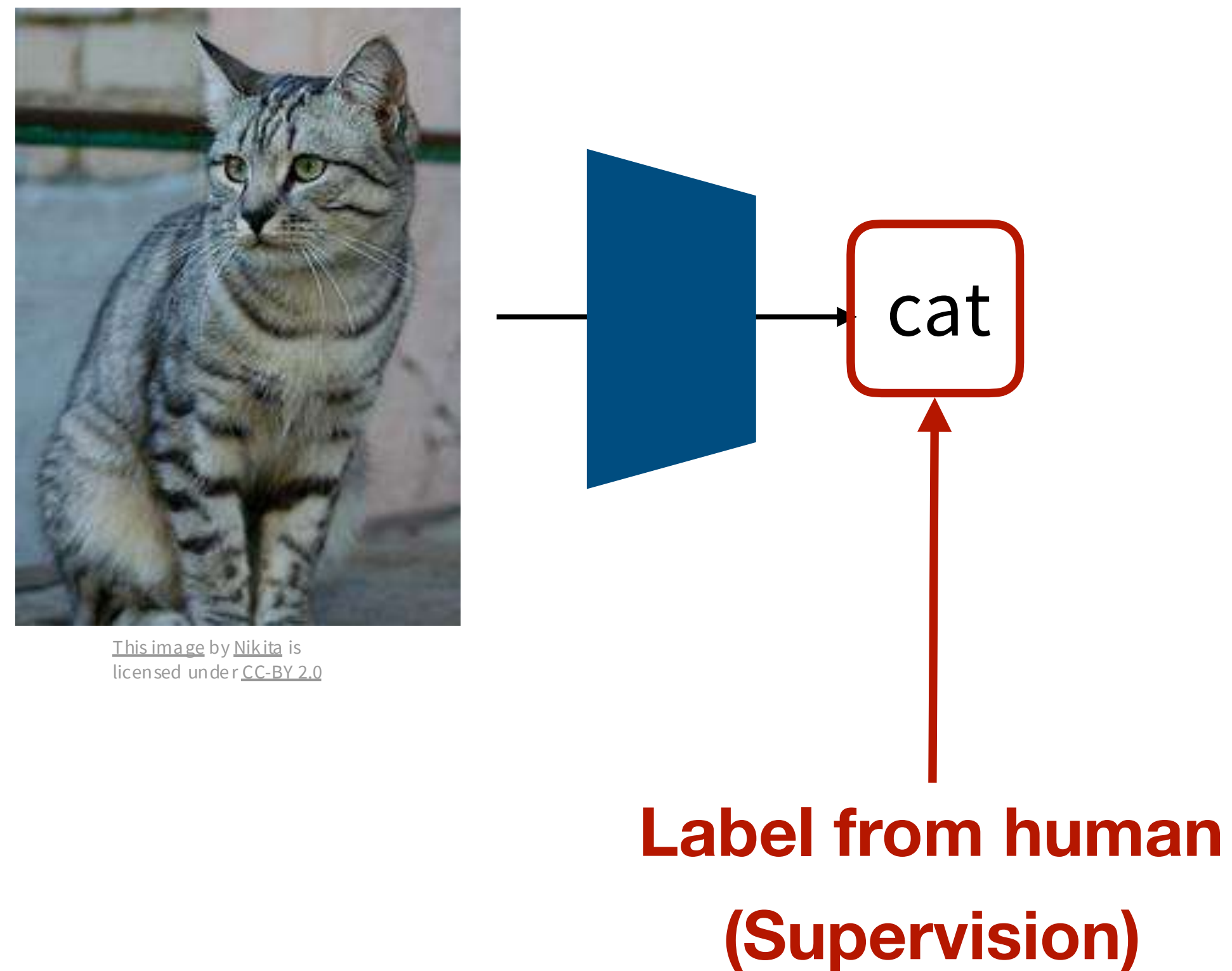
Rotation & Cut



Same

Foundation models

from supervised learning to self-supervised learning

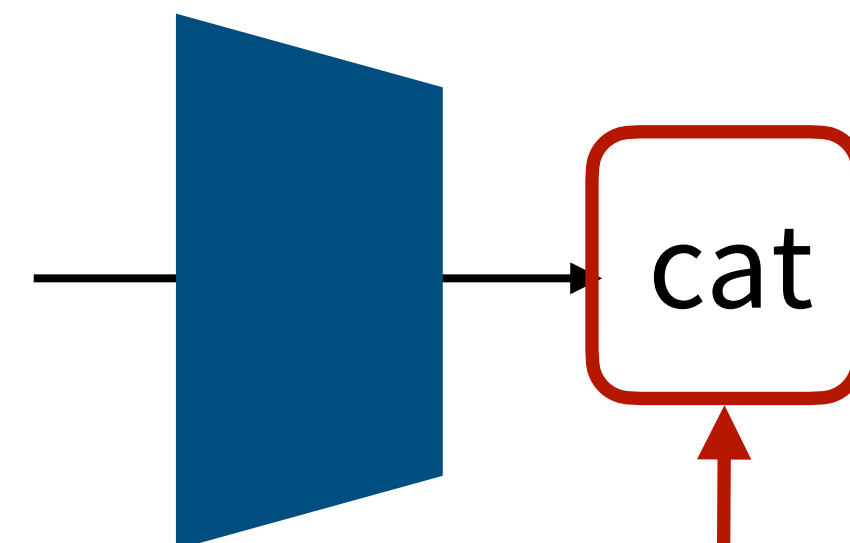


Foundation models

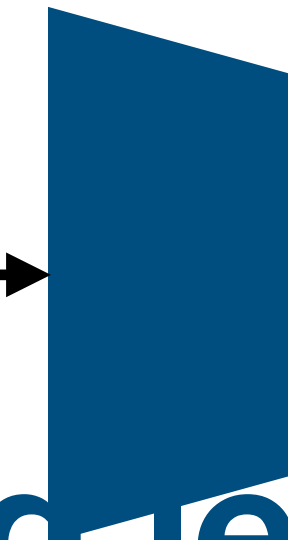
from supervised learning to self-supervised learning



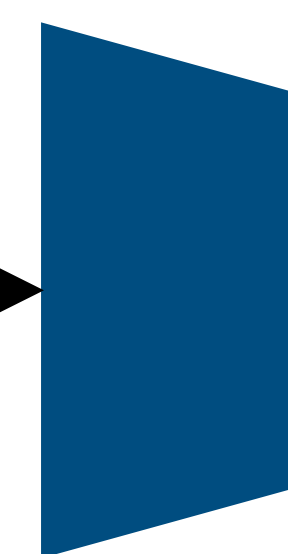
This image by Nikita is
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**Label from human
(Supervision)**



Dissimilar



Rotation & Cut

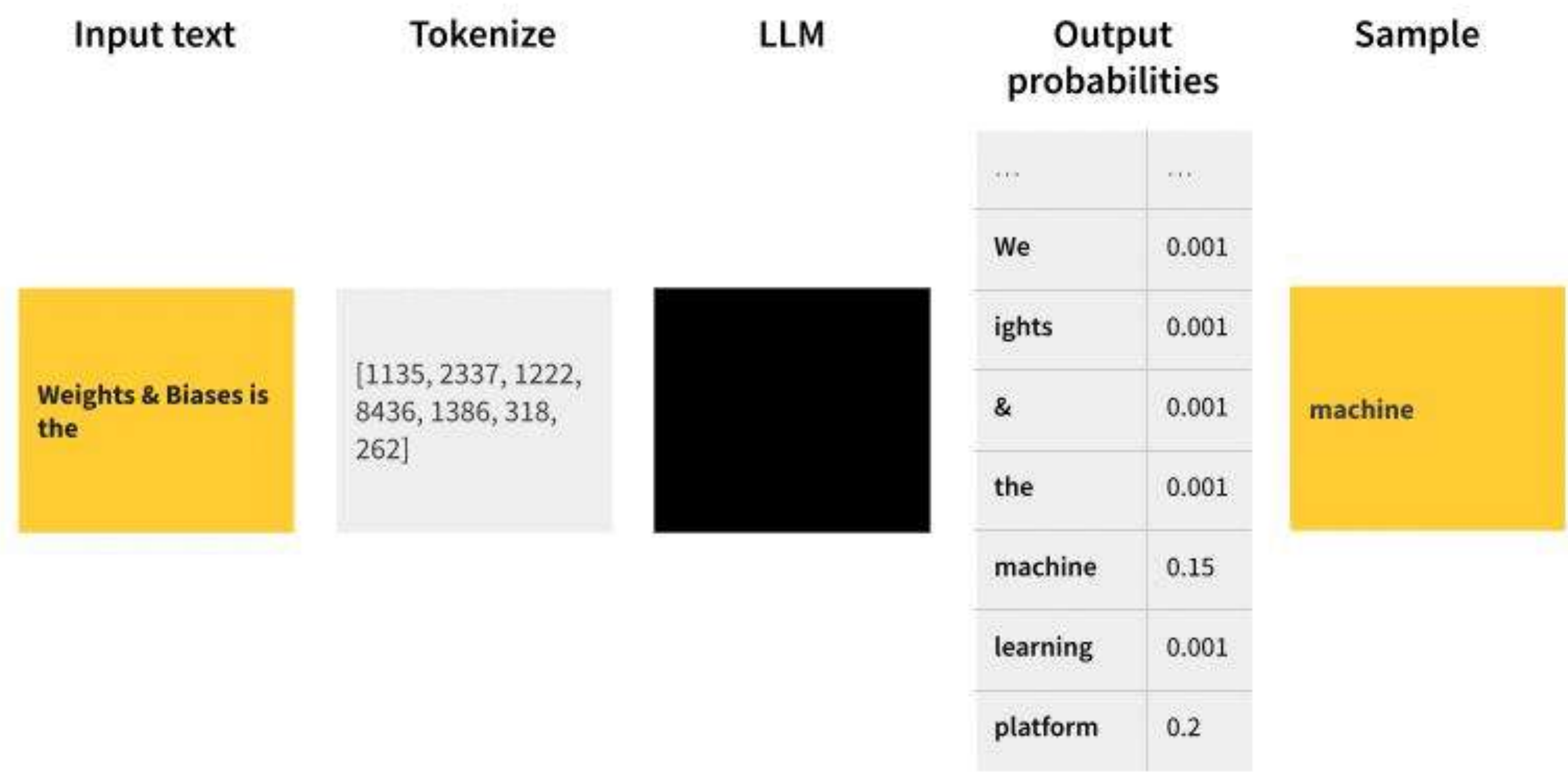


Similar

Large language Models (LLMs)

Autoregressive
Generation

Predict the next token

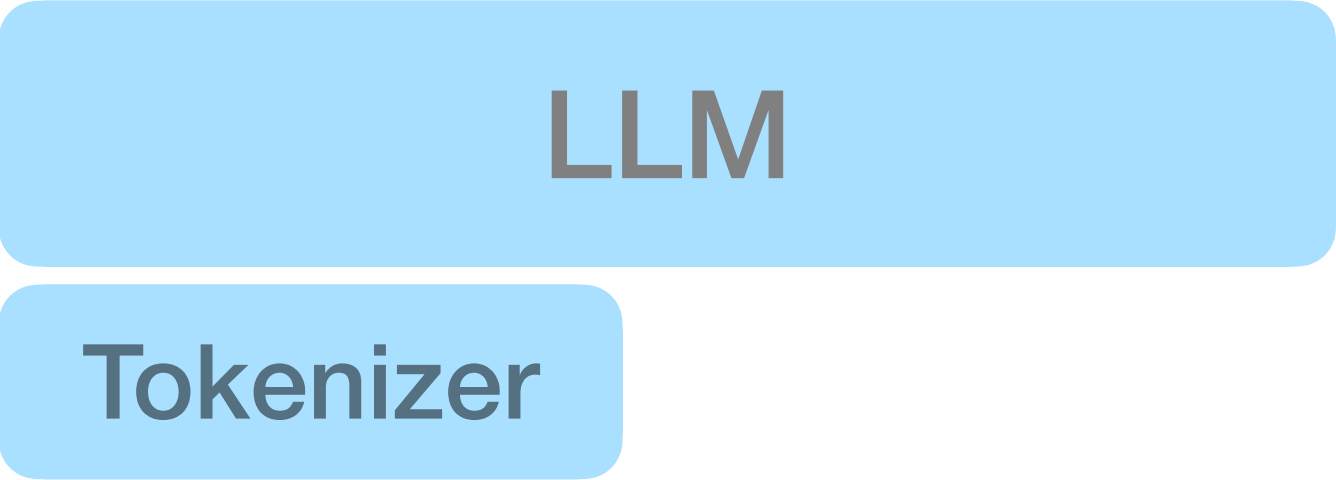


a probability distribution for the next token

Large language Models (LLMs)

Autoregressive
Generation

Predict the next token



Weights &
Biases is the

Input text

Tokenize

LLM

Output
probabilities

Sample

Weights & Biases is
the

[1135, 2337, 1222,
8436, 1386, 318,
262]



...	...
We	0.001
ights	0.001
&	0.001
the	0.001
machine	0.15
learning	0.001
platform	0.2

machine

Vision language Models (VLMs)

Text



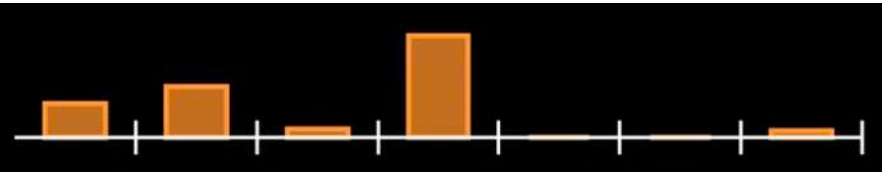
Prob on all
tokens

LLM

Tokenizer

Weights &
Biases is the

Text: Visual description / QA / ...



Prob on all
tokens

VLM

Tokenizer

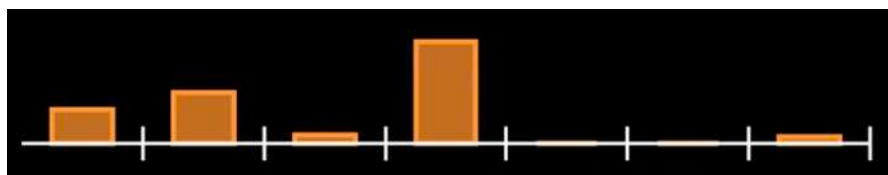
Vision Encoder

Weights &
Biases is the



MultiModal Language Models (MLMs)

Text



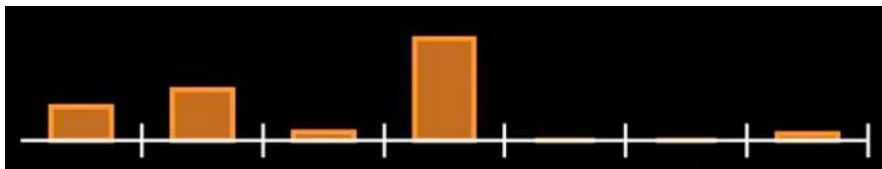
Prob on all
tokens

LLM

Tokenizer

Weights &
Biases is the

Text: Visual description / QA / ...



Prob on all
tokens

MLMs

Tokenizer

Weights &
Biases is the

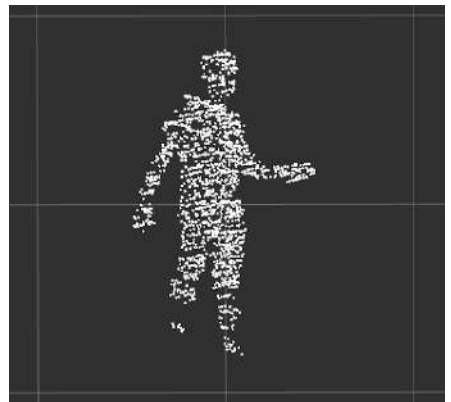
Vision Encoder



Audio Enc



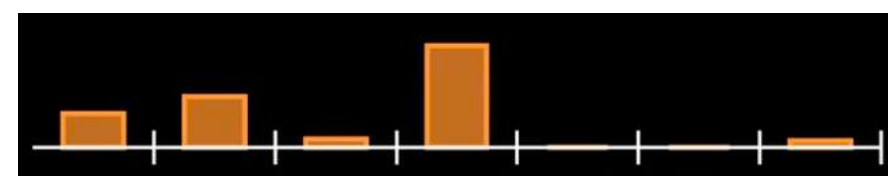
Lidar Enc



MultiModal Language Models (MLMs)

Understanding: multimodal data $\Rightarrow$ text!

Text



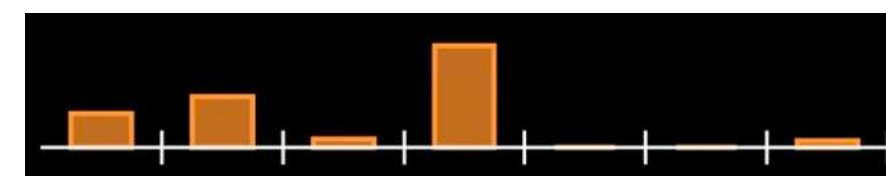
Prob on all
tokens

LLM

Tokenizer

Weights &
Biases is the

Text: Visual description / QA / ...



Prob on all
tokens

MLMs

Tokenizer

Weights &
Biases is the

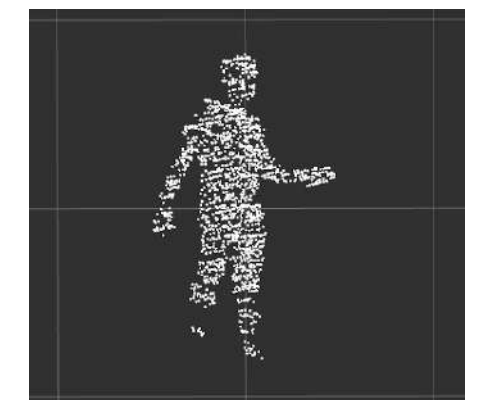
Vision Encoder



Audio Enc

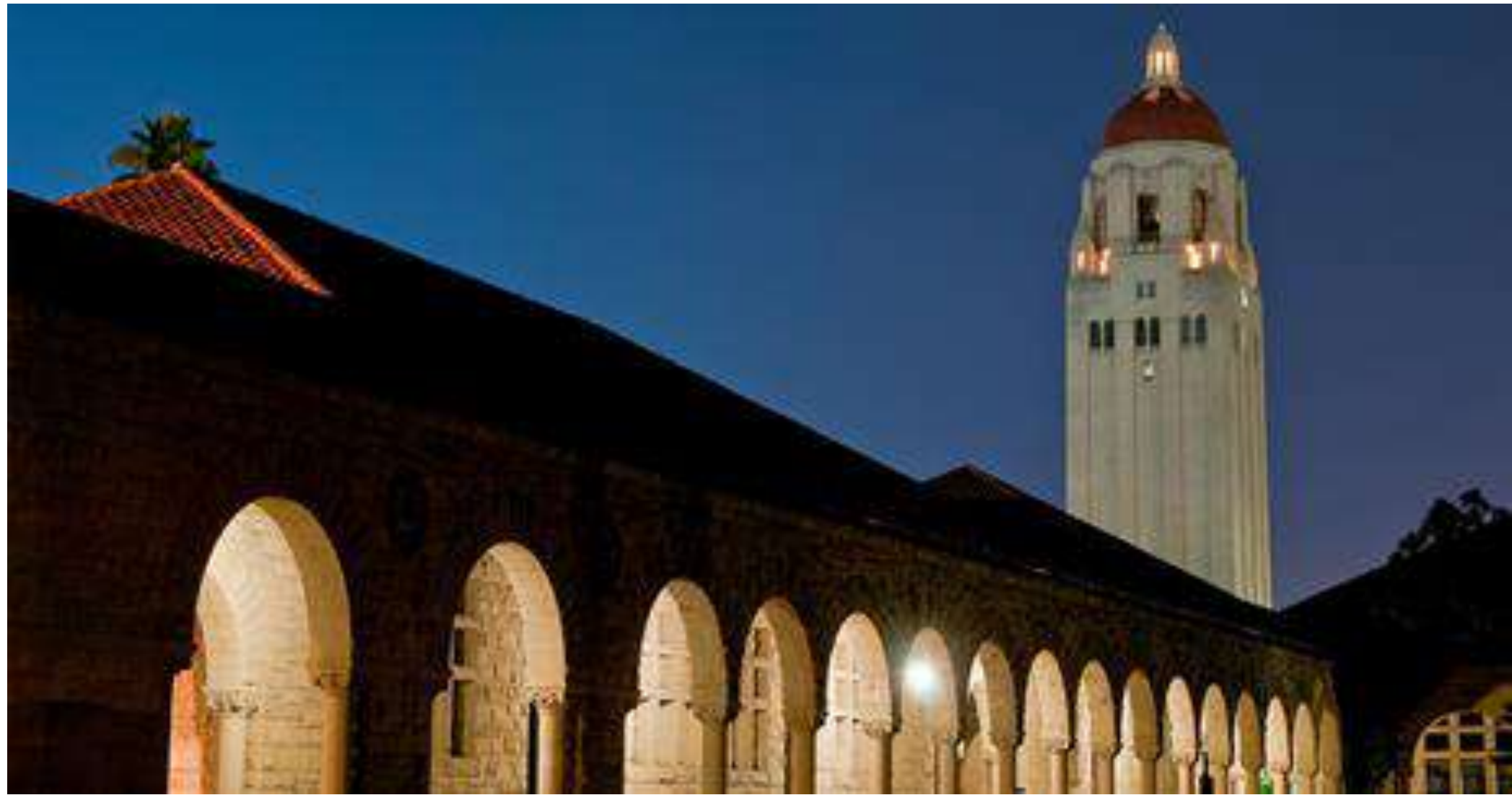


Lidar Enc



Generative Models

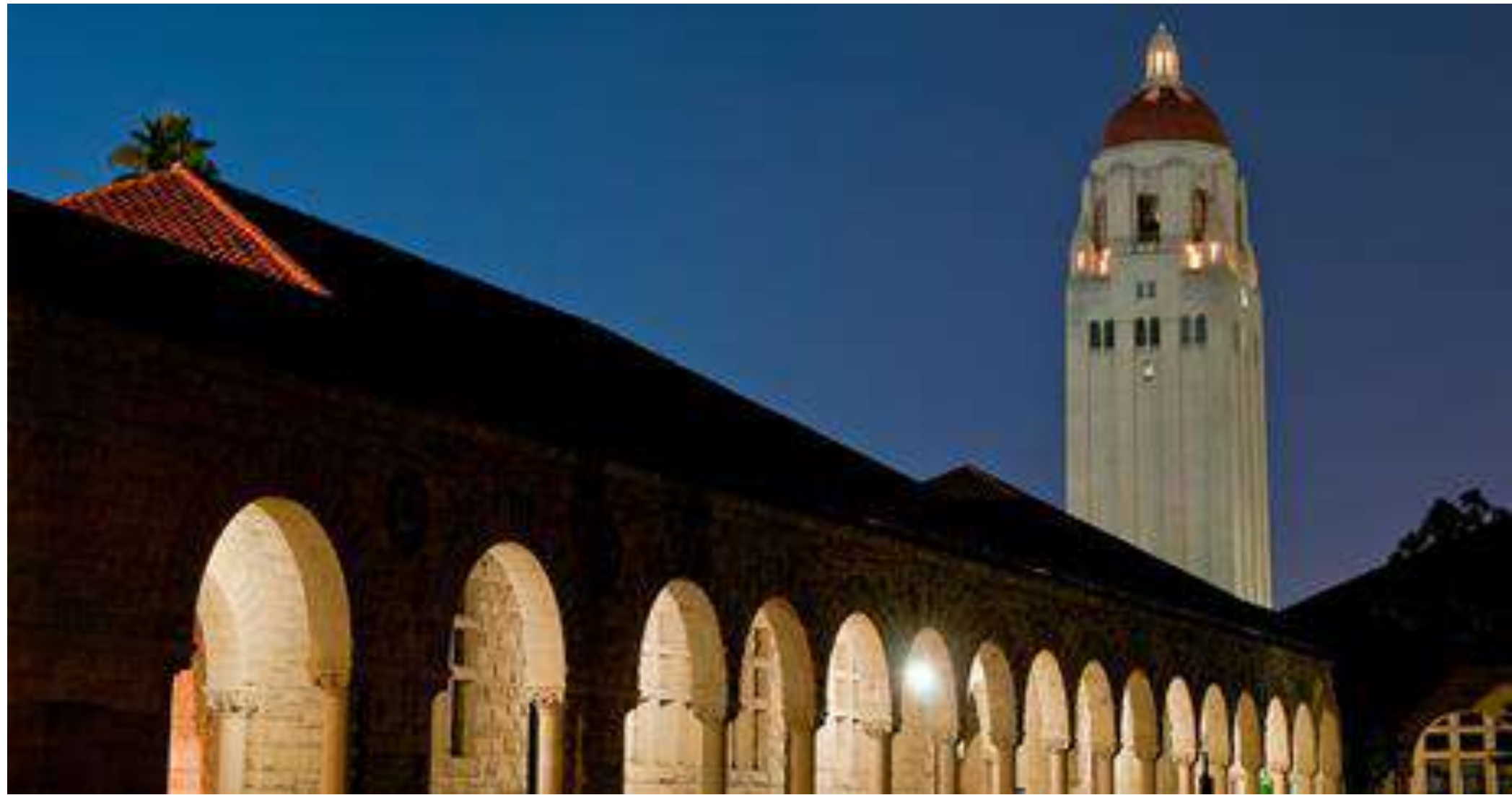
Generative Modeling: Style Transfer



Your photo

Generative Models

Generative Modeling: Style Transfer



Your photo



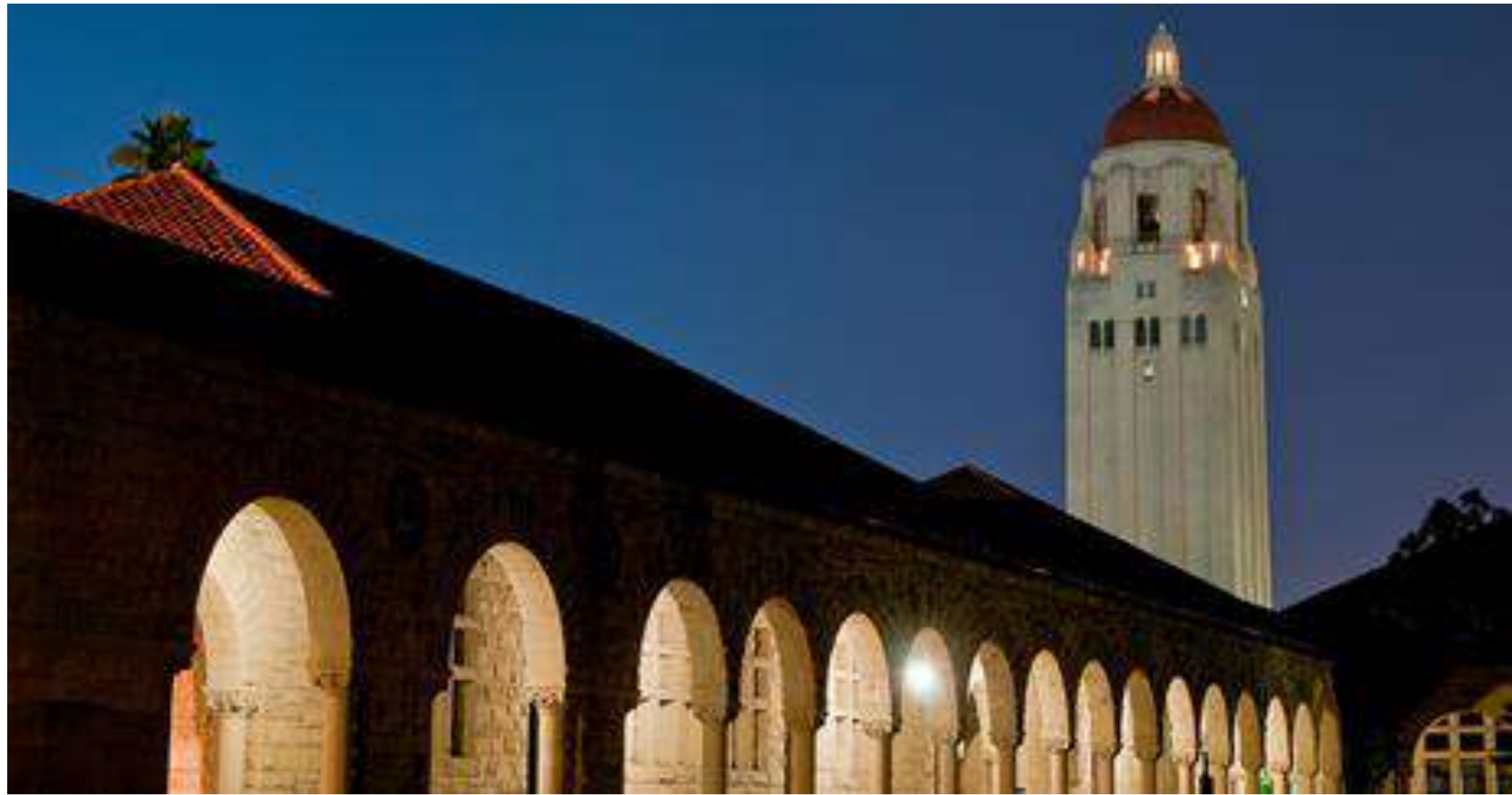
Van Gogh Style



Your photo + Van Gogh Style?

Generative Models

Generative Modeling: Style Transfer



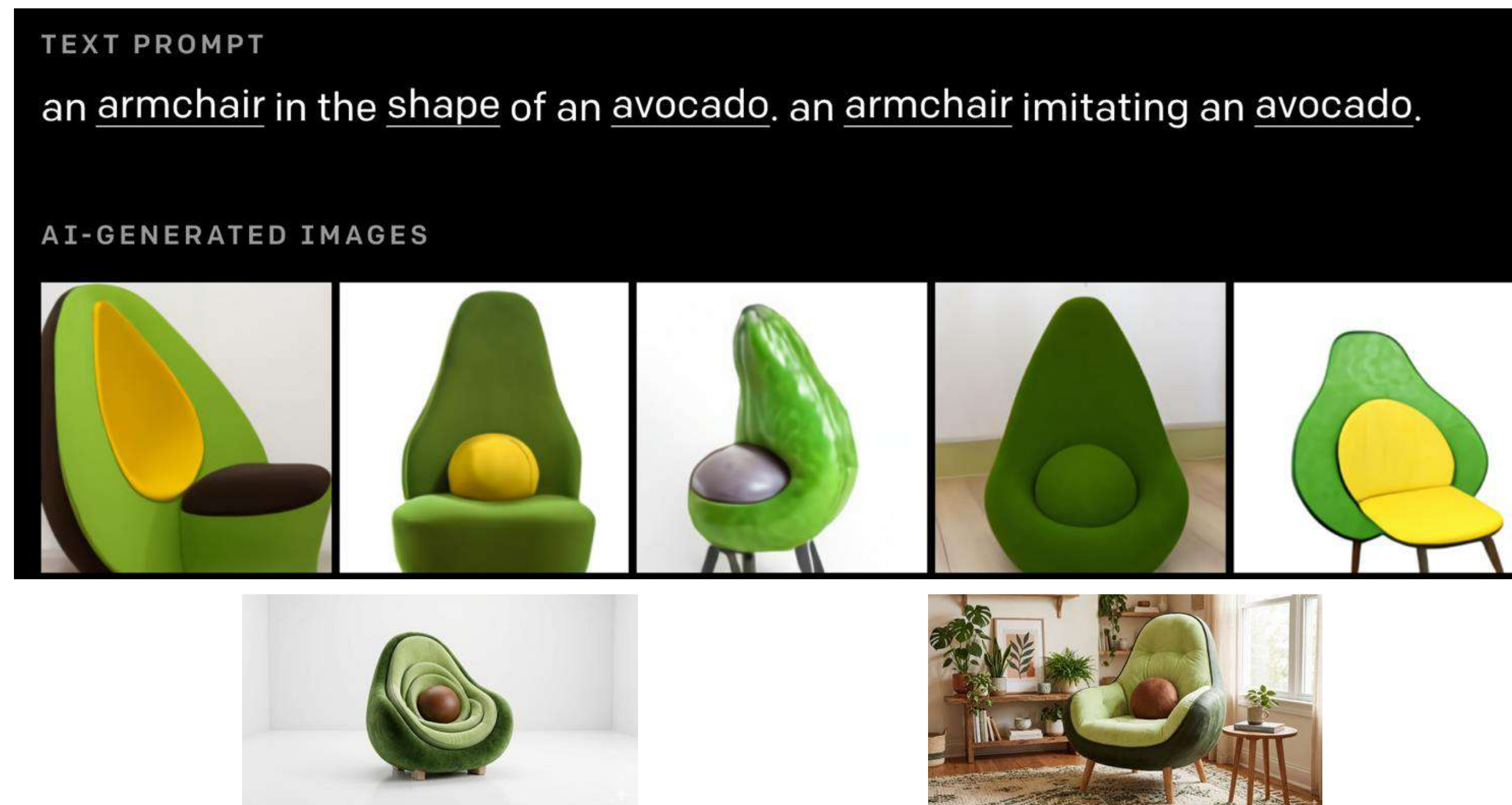
Your photo



Your photo + Van Gogh Style

Generative Models: text $\Rightarrow$ multimodal data!

Generative Modeling: Text-to-image generation



Slide inspiration: Justin Johnson

Ramesh et al, "DALL·E: Creating Images from Text", 2021. <https://openai.com/blog/dall-e/>
Remaining images generated by GPT-4o and Imagen 4

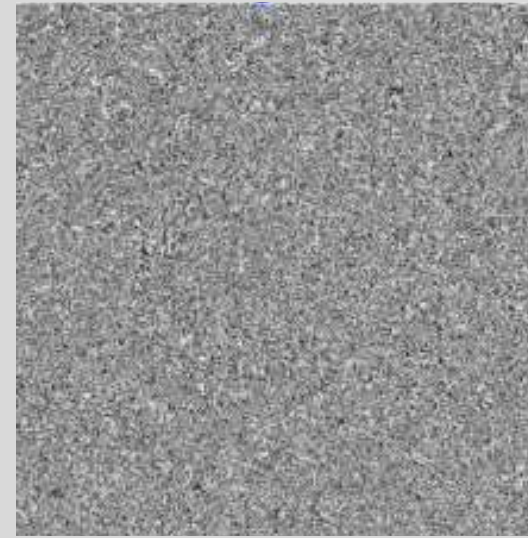
Realistic or compositional images

Generative Models: text $\Rightarrow$ multimodal data!

Generative Modeling: Text-to-image generation

Input

**Gaussian
noise**



Text guidance:

A small orange-and-white
tiger lies on the grass.

Output

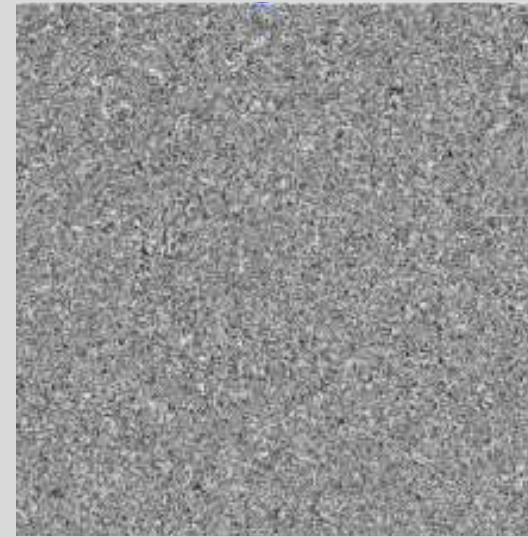


Generative Models: text $\Rightarrow$ multimodal data!

Generative Modeling: Text-to-image generation

Input

Gaussian
noise



Text guidance:

A small orange-and-white
tiger lies on the grass.

Output



Low-frequency
information

High-frequency
details

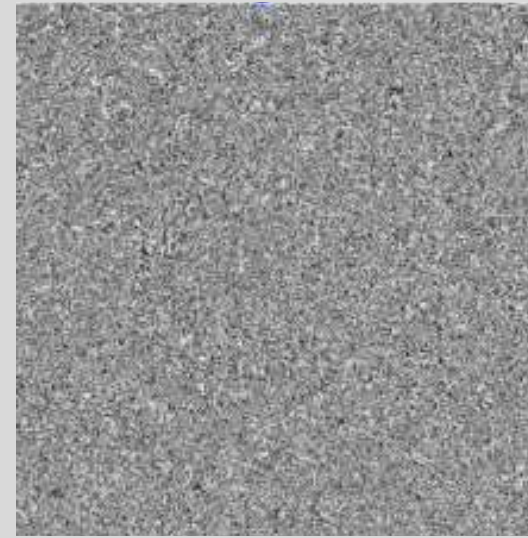
When humans create a painting:
Global layout / Rough color blocks
→ Basic shapes → Structural
sketch → Refined structure →
Colors and details → Final painting

Generative Models: text $\Rightarrow$ multimodal data!

Generative Modeling: Text-to-image generation

Input

Gaussian
noise



Text guidance:

A small orange-and-white
tiger lies on the grass.

Output



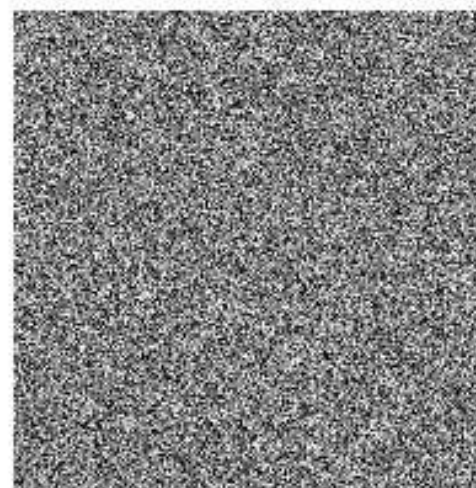
T2I Gen.

Multiple time-step denoising process

Diffusion/Flow Models

Reverse (Denoising)

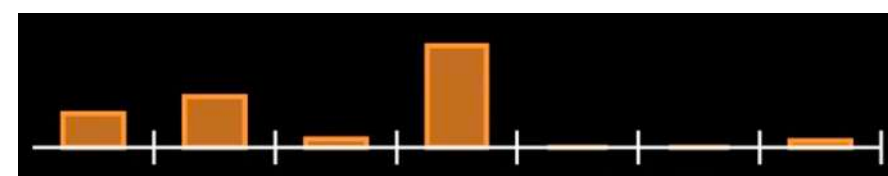
t = 1.00



MultiModal Language Models (MLMs)

Understanding + Generation:
multimodal data $\Rightarrow$ multimodal data!

Text



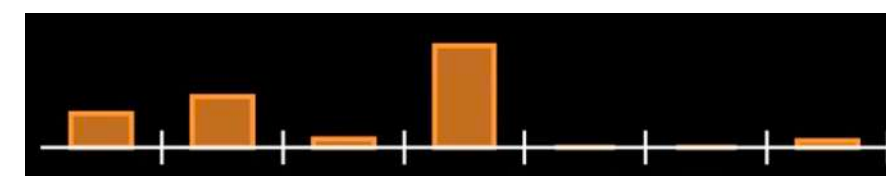
Prob on all
tokens

LLM

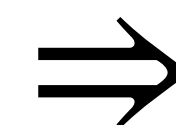
Tokenizer

Weights &
Biases is the

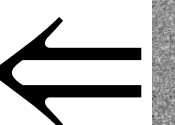
Text: Visual description / QA / ...



Prob on all
tokens



Diffusion Decoder



MLMs

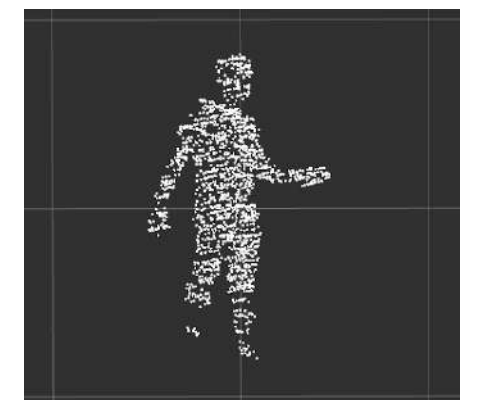
Tokenizer

Vision Encoder

Audio Enc

Lidar Enc

Generate an image: a
cat (in the left figure)
is skiing in the snow



Geometry, 3D vision and spatial understanding

3D Shape Representations: Images are 2D projections of a 3D world.

Pixel is a single flat square on a 2D screen.

Input image



2D object recognition



3D object meshes

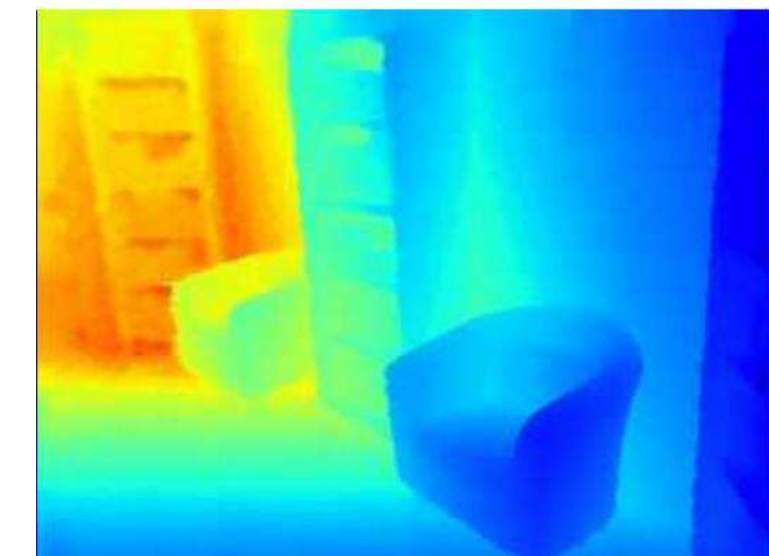
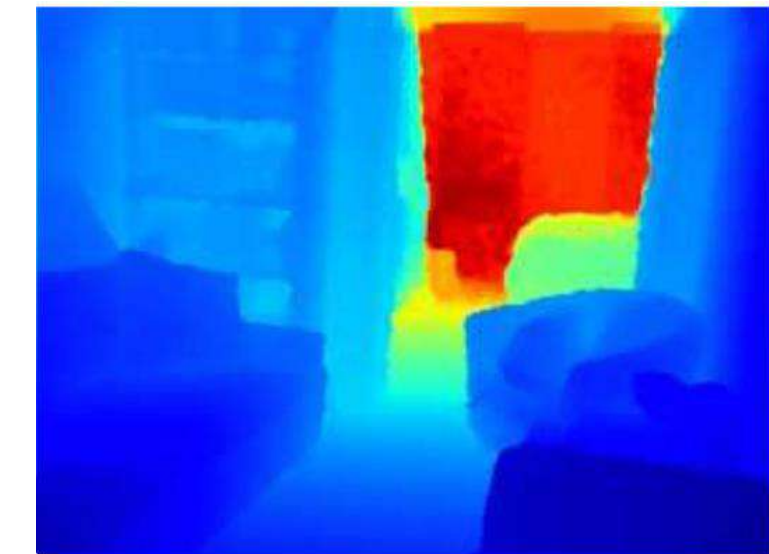
A grid of triangles or polygons used to map out digital shapes.

Mesh R-CNN pipeline



3D object voxels

A tiny cube / 3D volume element



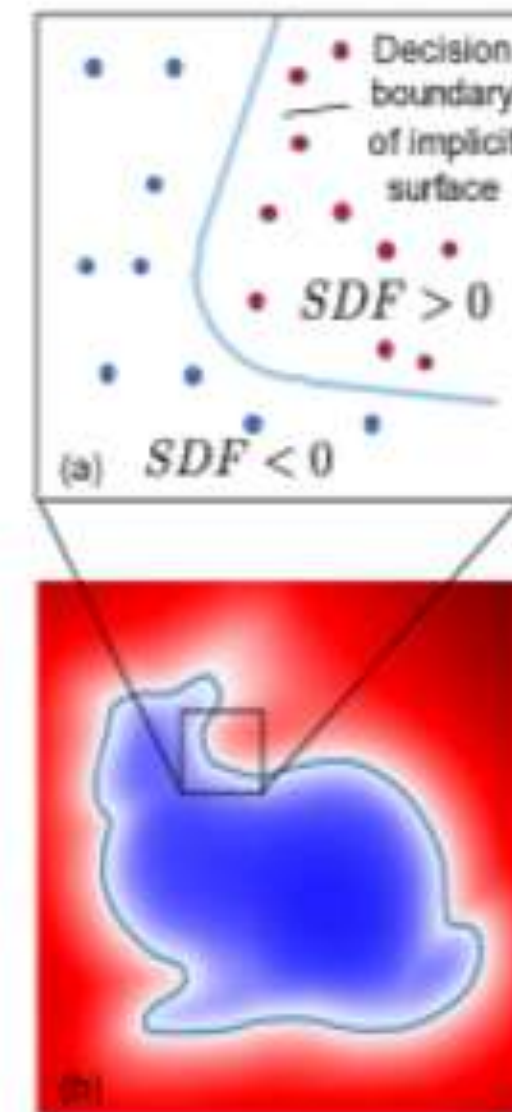
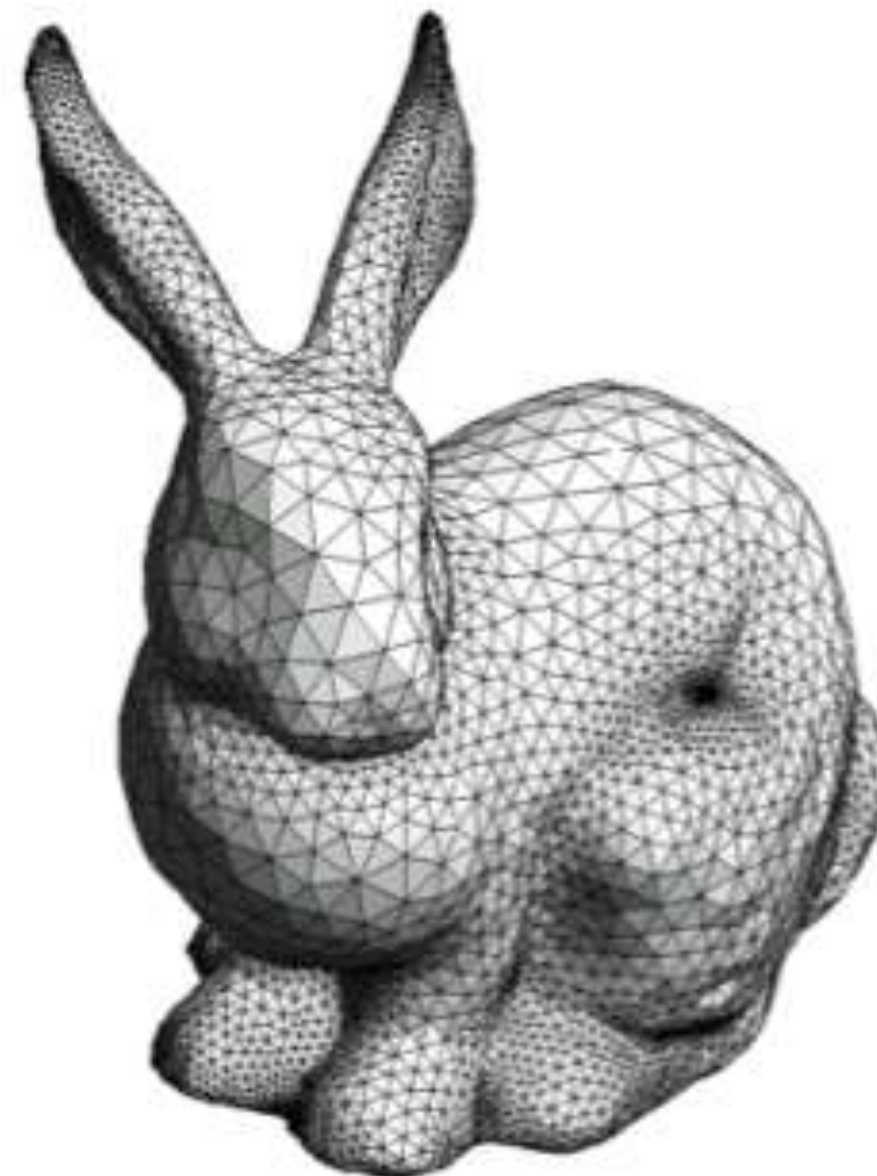
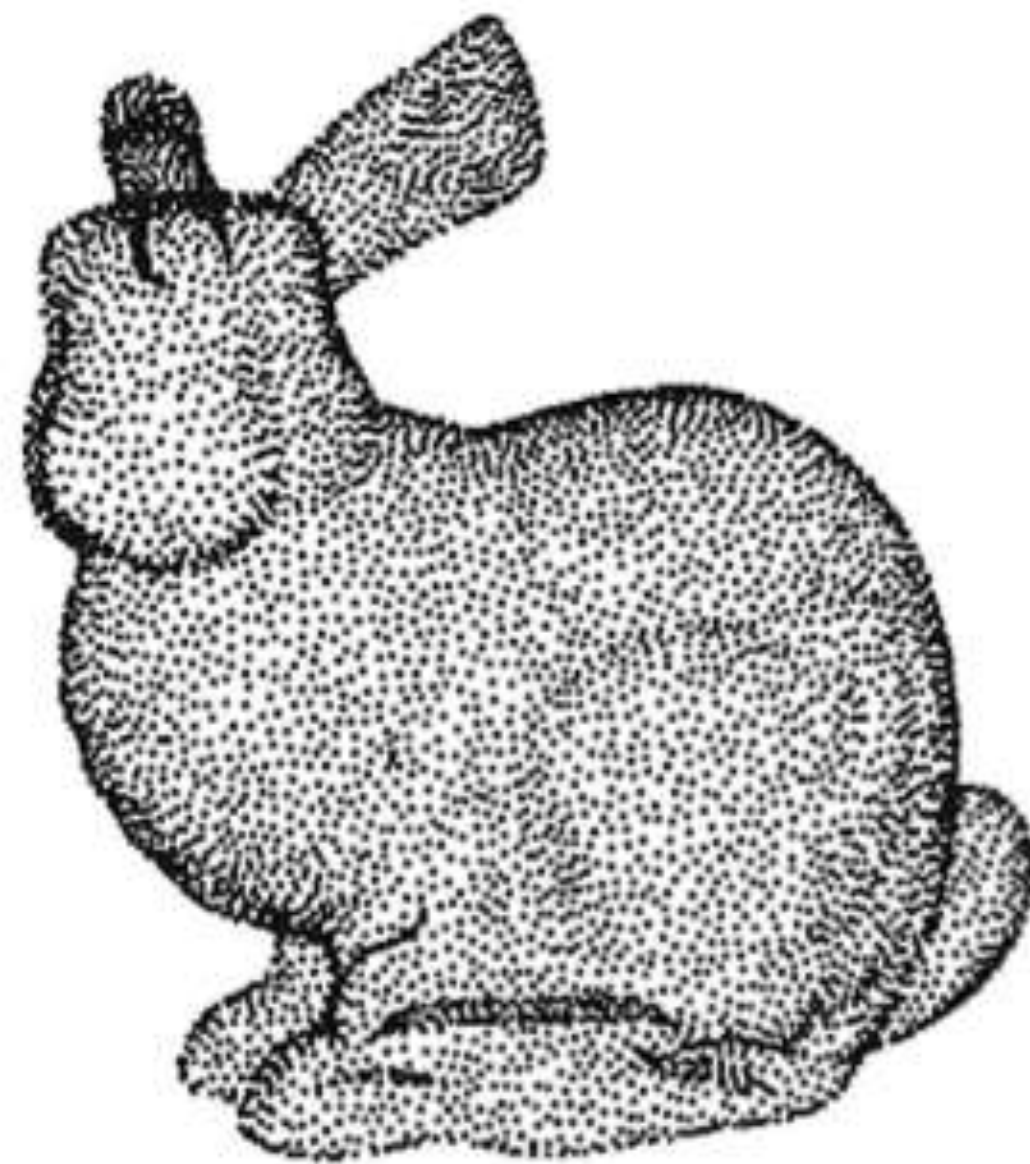
RGB Image: $3 \times H \times W$ Depth Map: $H \times W$

Depth Map (2.5D)

visible surfaces from one viewpoint

Geometry, 3D vision and spatial understanding

3D Shape Representations



Geometry, 3D vision and spatial understanding

3D Shape Representations + Attribute Representations (color, materials, ...)

(Geometry)

(Physical properties)



Geometry, 3D vision and spatial understanding

3D Shape Representations + Attribute Representations (Color, materials)

(Geometry)

(Physical properties)



Technical Landscape

Recognition & Localization & Motion

Classification

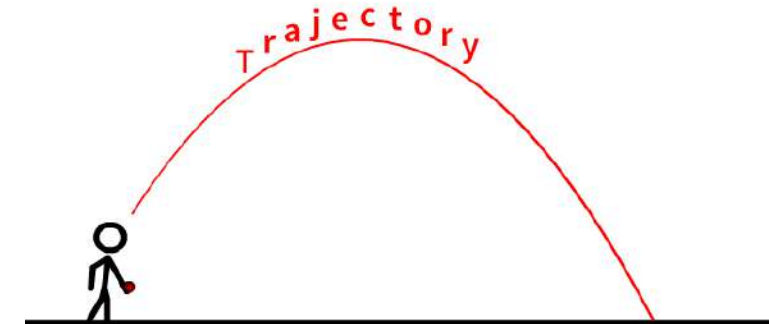


Apple, **Burro**, Cat, Horse ...

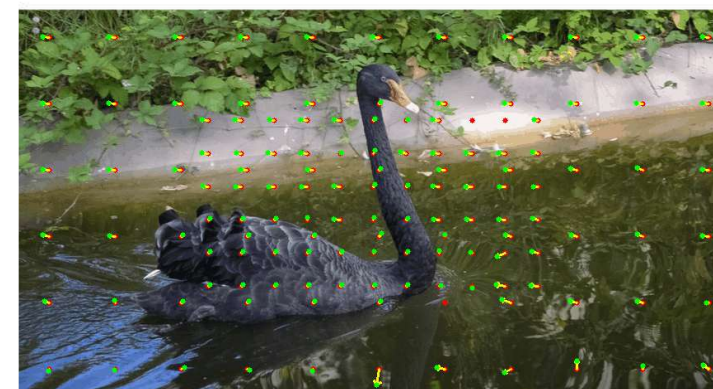
Object Detection



Segmentation



Point Tracking



Object Tracking



Foundation Models



A cat

A dog

A nuclear submarine



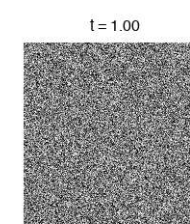
CLIP (Contrastive Language-Image Pre-training)

Predict the next token

Input text	Tokenize	LLM	Output probabilities	Sample
Weights & Biases is the	[1135, 2337, 1222, 8436, 1386, 318, 262]		...	...
			We	0.001
			ights	0.001
			&	0.001
			the	0.001
			machine	0.15
			learning	0.001
			platform	0.2
				machine

LLMs

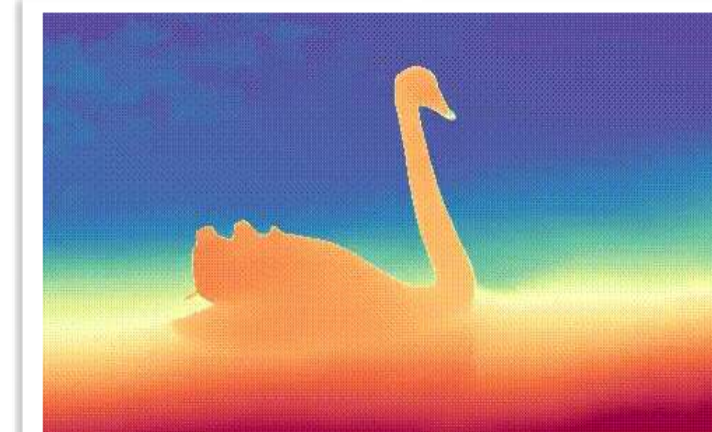
Reverse (Denoising)



Diffusion Models

Geometry & 3D Vision

Depth Estimation



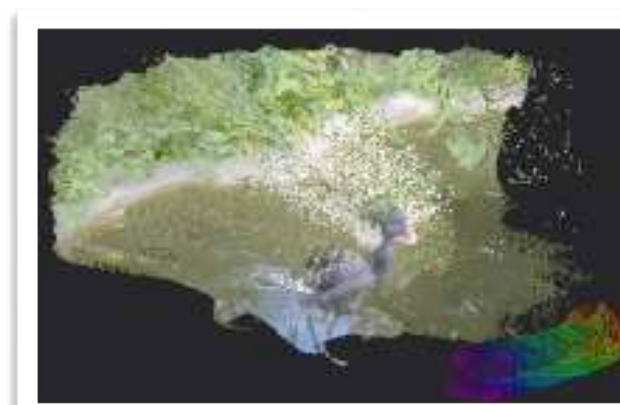
Multi-view Matching



3D Object Reconstruction



3D Scene Reconstruction



What / where is in the world?

What is the 3D structure of the world?

How can computer vision connect with applications in engineering?

CSCI 598H Course Overview



Weeks 2-4

Weeks 5-11

Weeks 12-16

Module 1

Deep Learning Basics

- Linear Classifier and Optimization
- CNNs and BackPropagation
- RNNs, Attention, Transformers

Module 2

Modern Visual Intelligence

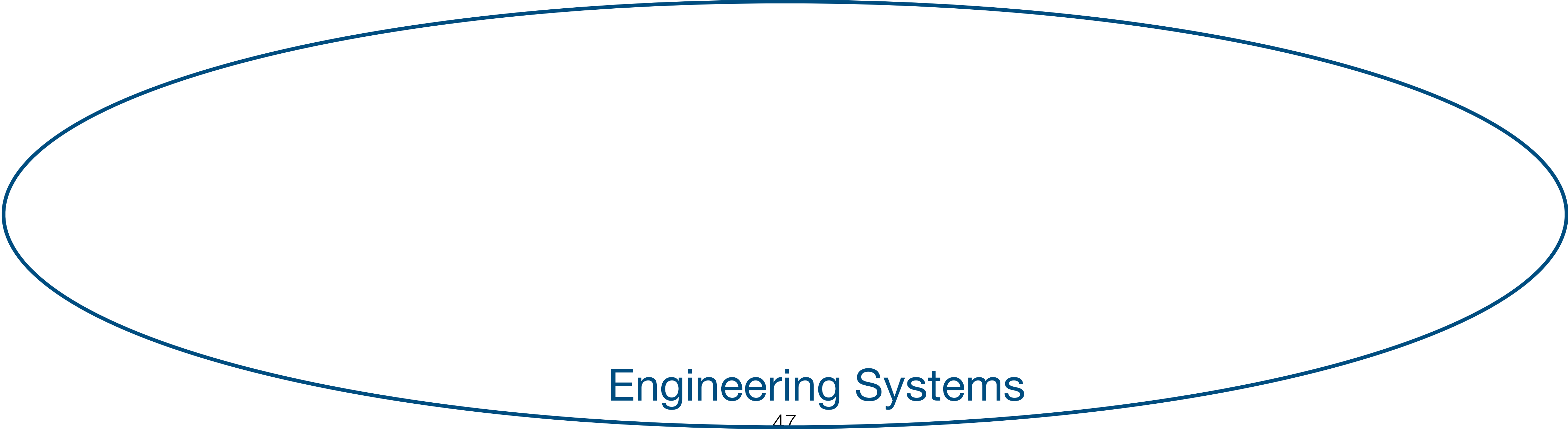
- Perceiving and understanding the visual world
- Generative models and foundation models
- Geometry, 3D vision and spatial understanding

Module 3

Engineering Systems

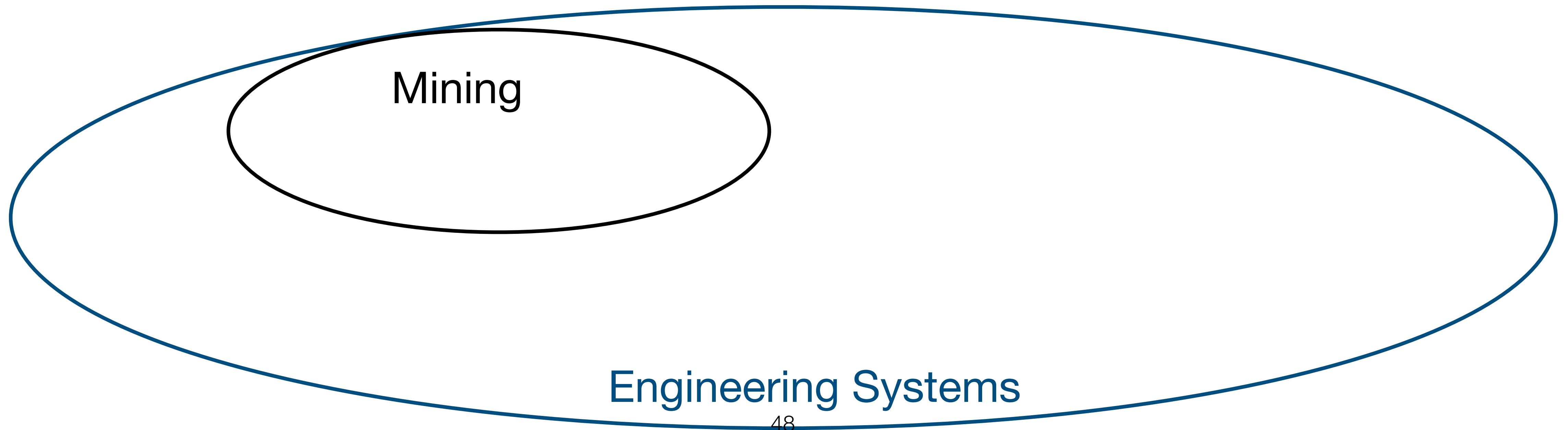
- Robotics
- Earth science
- Medical imaging
- Energy

And there is a lot we don't know how to do

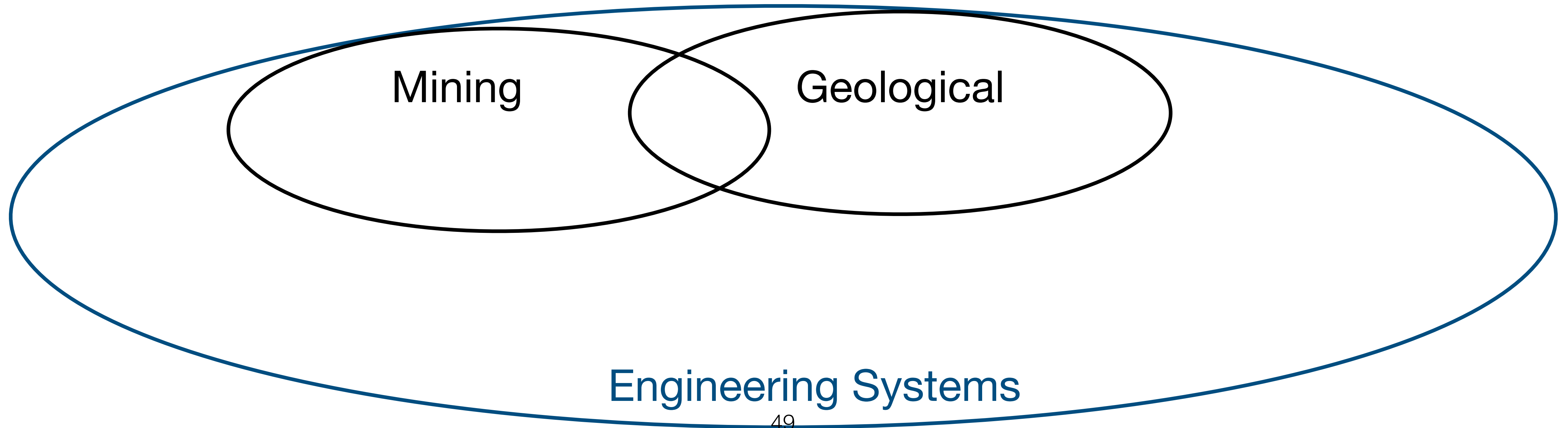


Engineering Systems

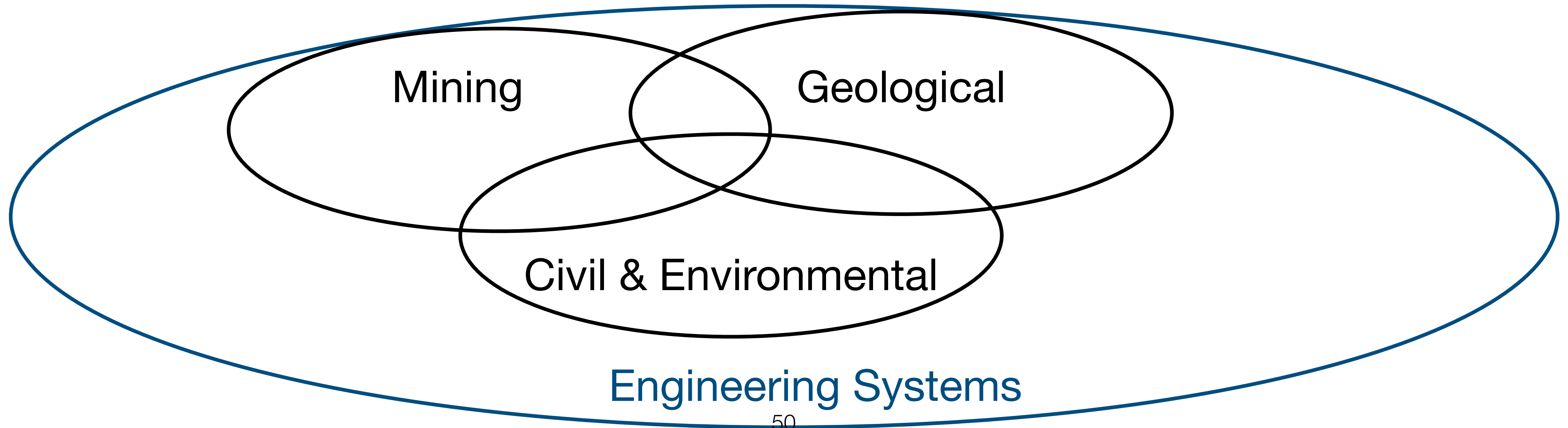
Course Scope



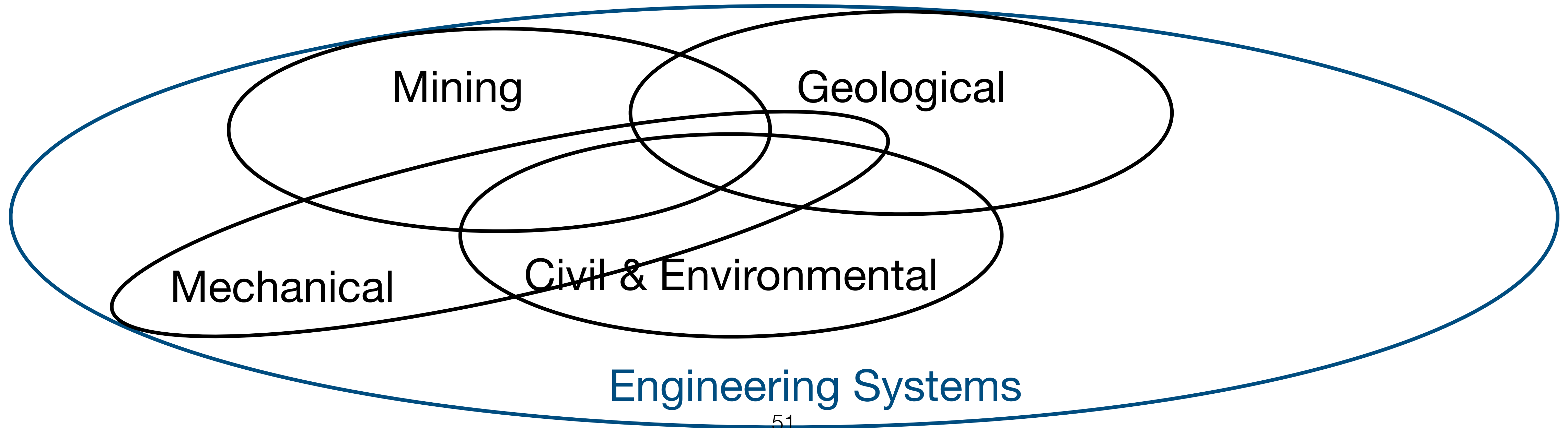
Course Scope



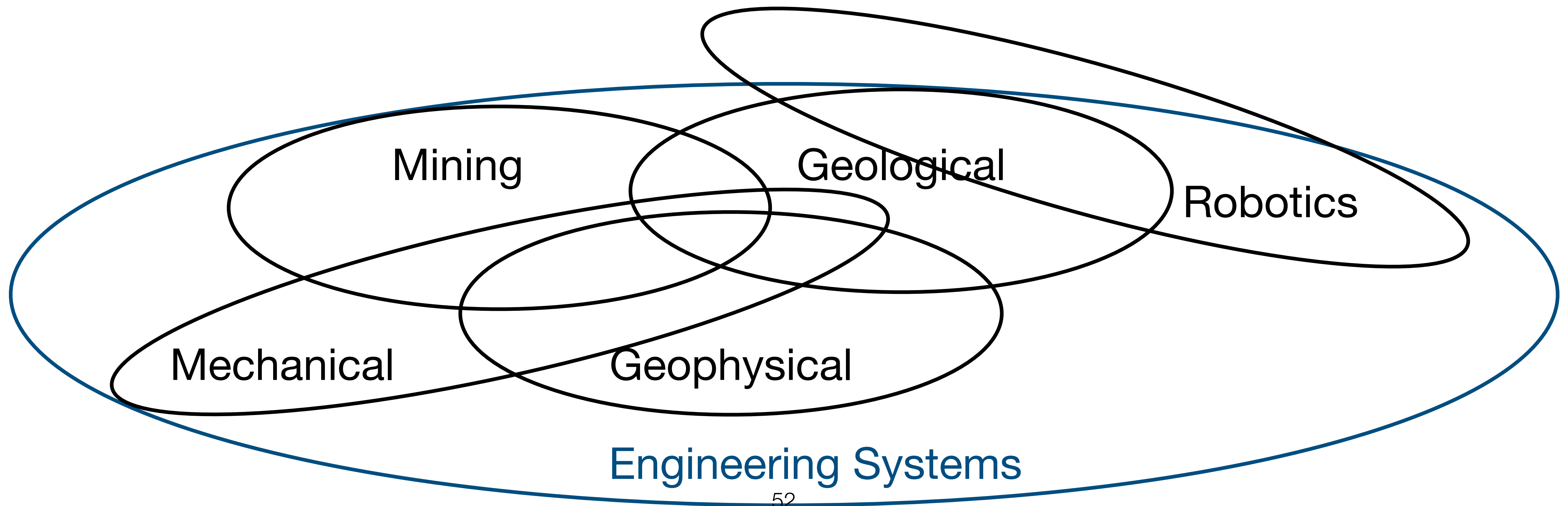
Course Scope



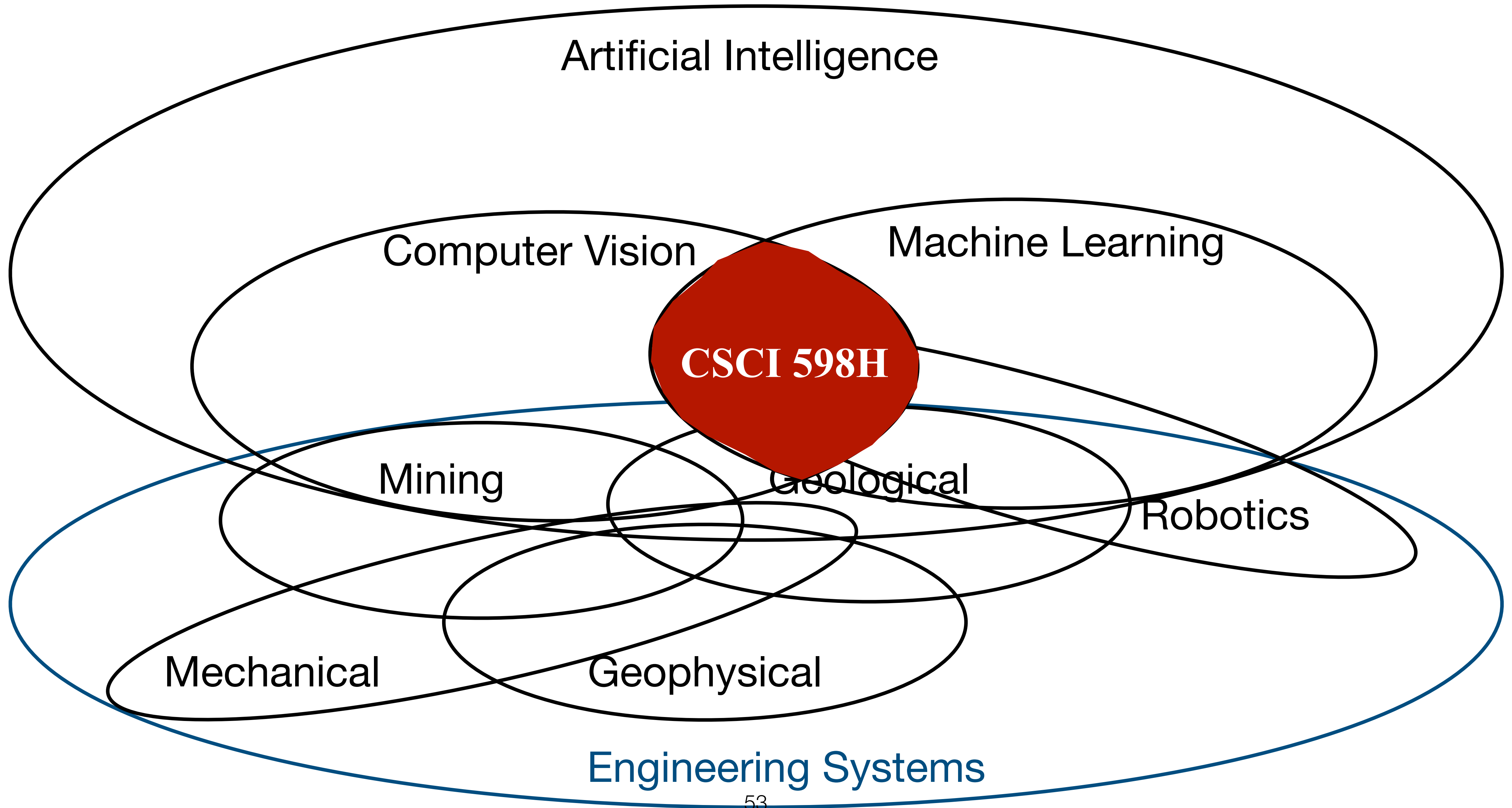
Course Scope



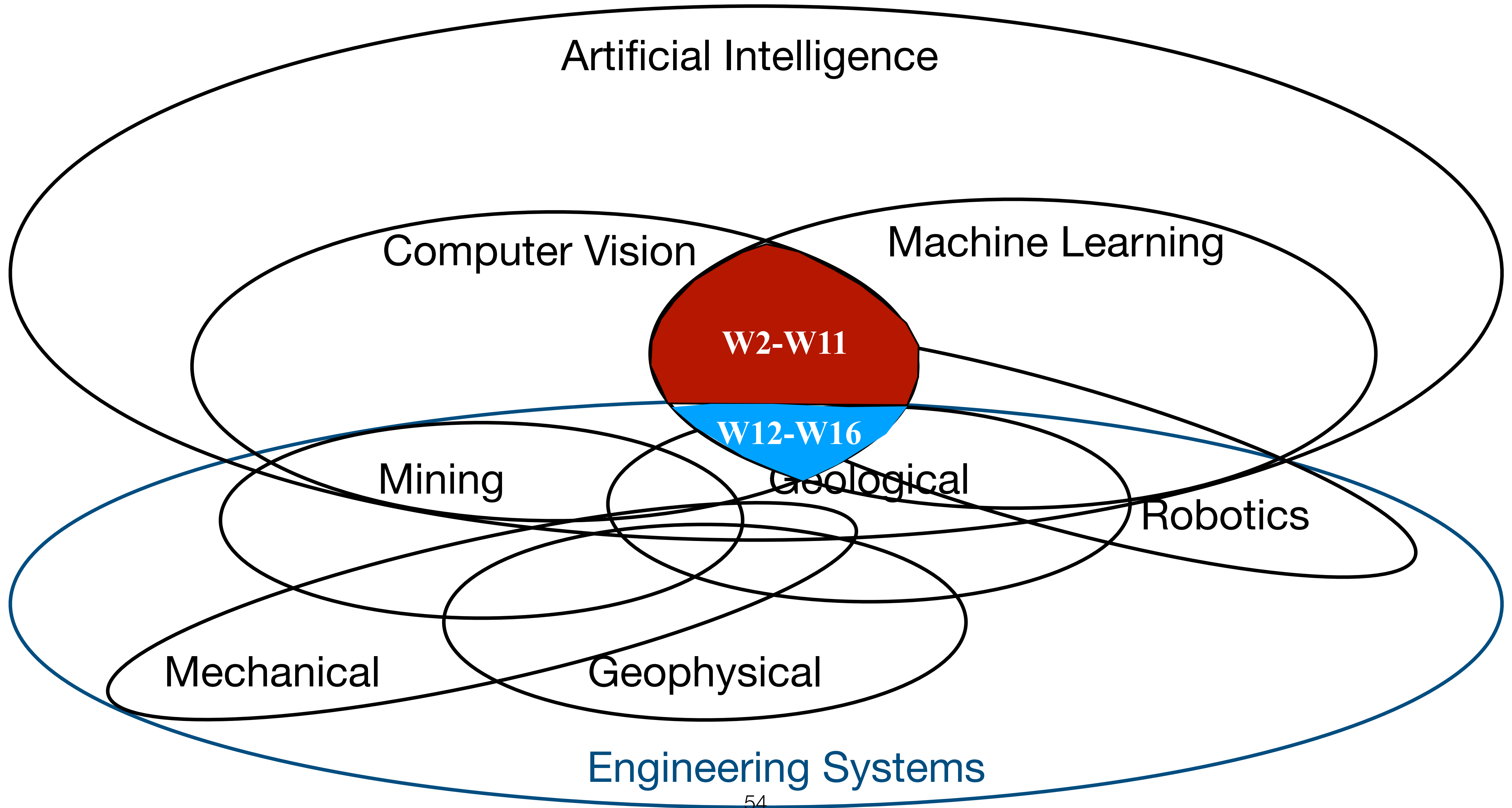
Course Scope



Course Scope

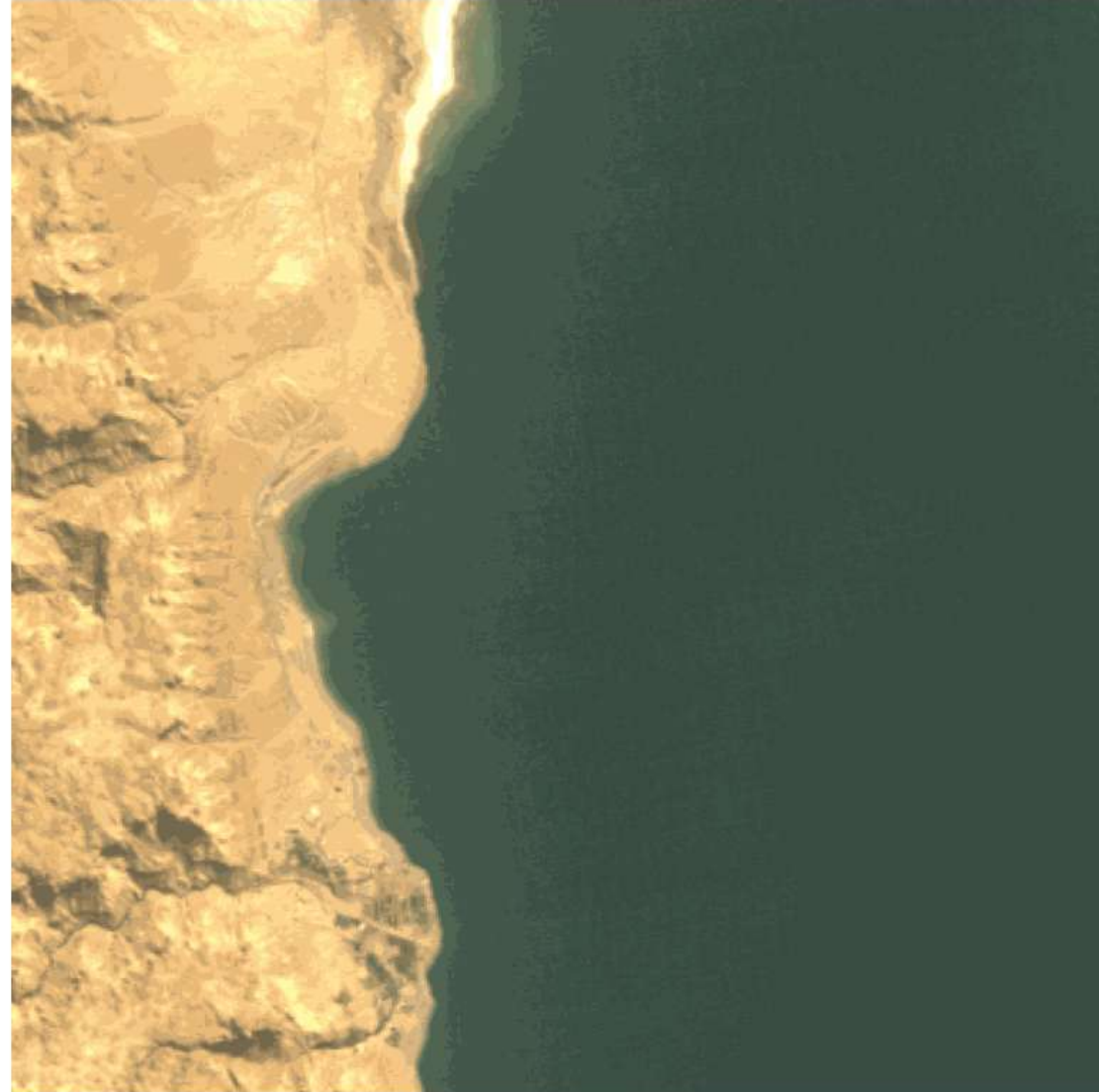


Course Scope

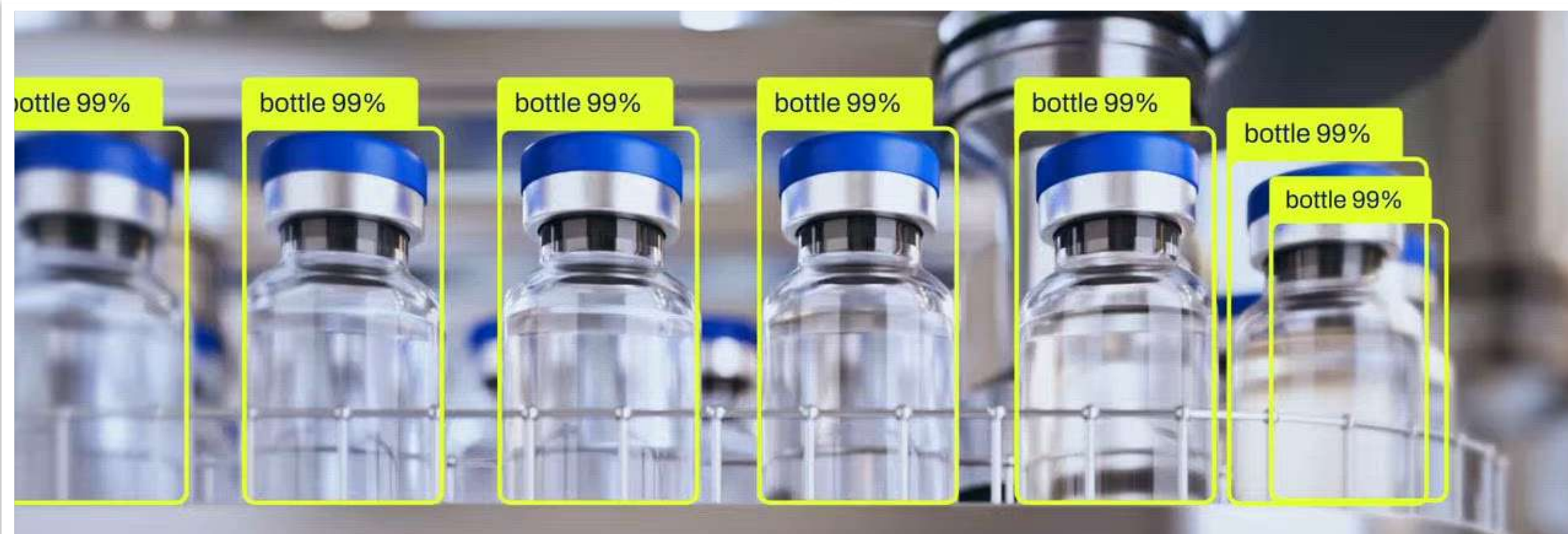
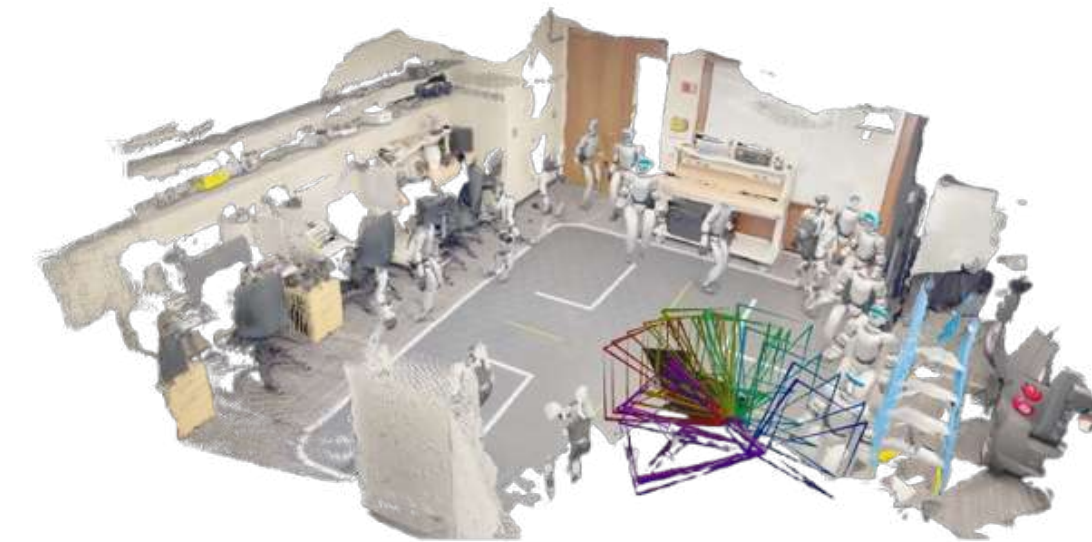
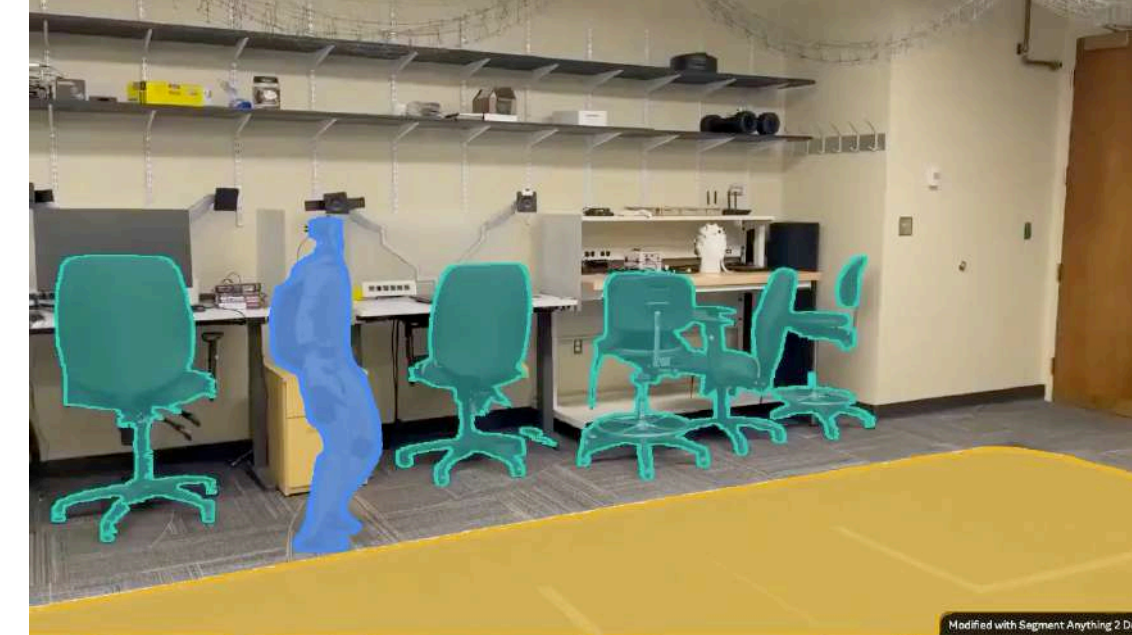


Engineering Applications

Year: 1984



[Link](#) The Dead Sea drying up



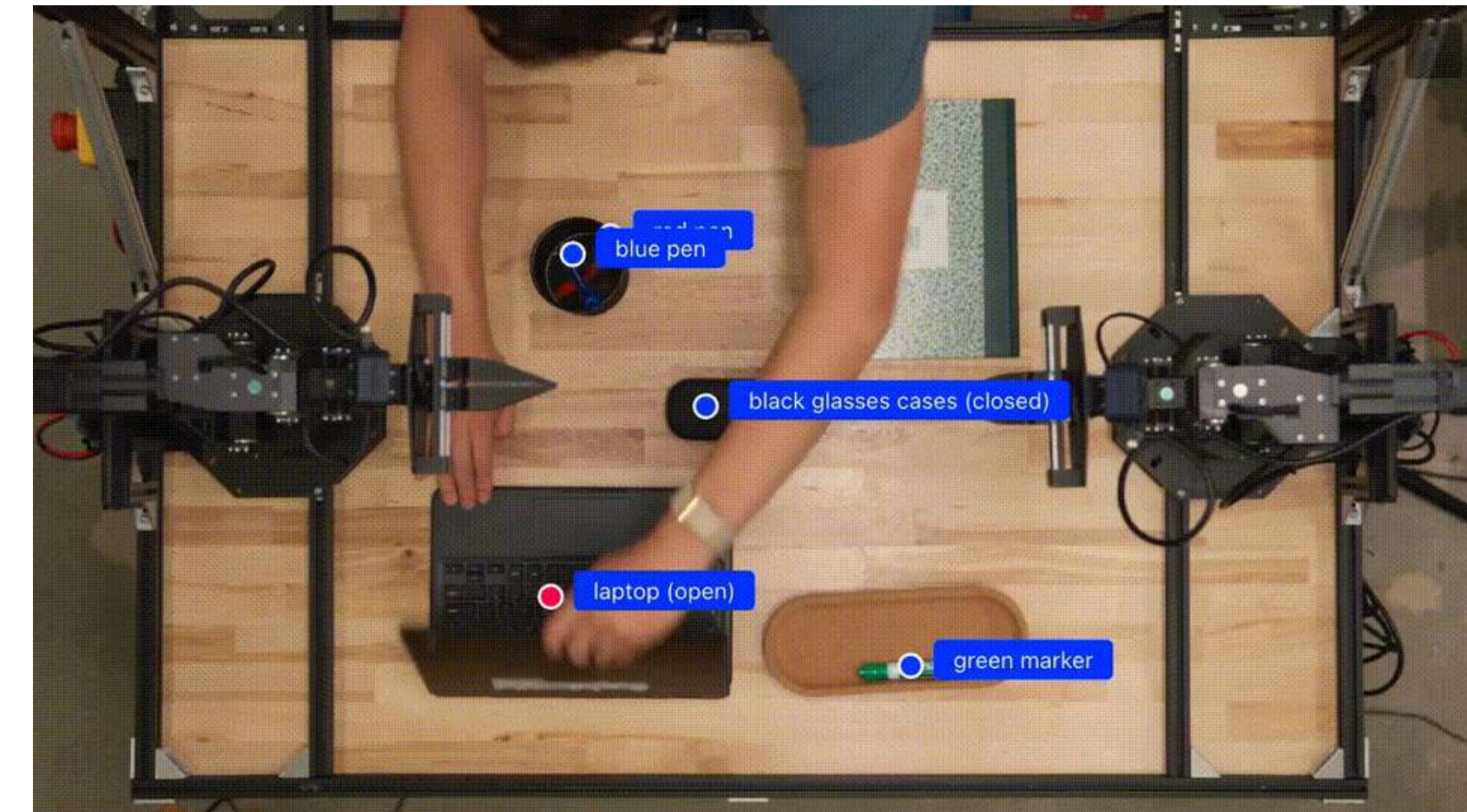
[Link](#)

Engineering Applications



<https://www.pi.website/blog/pi05>

Pi (VLA Model)



<https://deepmind.google/models/gemini-robotics/gemini-robotics-er/>

Gemini - Robotics

Today's agenda

- A brief history of computer vision (CV)
- CSCI 598H course overview
- **CSCI 598H course administration**

Table 1: 16-Week Course Schedule

	Module	Week Dates	Lecture Topic	Core Technical Focus & Key Algorithms
Career Days	Module 1: CV & DL Basics	W1 8/25, 8/27	Introduction	Introduction and Course Overview
		W2 9/1, 9/3	Basics of Optimization	Brief history; Linear Classifier; Regularization
		W3 9/8, 9/10	CNN & Backpropagation	Convolutional neural networks (CNNs); Backpropagation; SGD
		W4 9/15 , 9/17	RNNs & Attention & Transformers	Attention mechanisms; Transformers & Vision Transformers
Fall Break	Module 2: Advances in CV	W5 9/22, 9/24	Recognition & Localiza-tion	Classification, Detection and Segmentation (ResNet, YOLO, Fast R-CNN, Mask R-CNN, ViT, DETR)
		W6 9/29, 10/1	Video Understanding	Motion estimation with optical flow, temporal modeling with 3D CNNs, RNNs, and Video Vision Transformers;
		W7 10/6, 10/8	Large-Scale Distributed Training	GPUs & Cluster, Data/Context/Tensor/Pipeline Parallelism, How to train on GPUs
		W8 10/13 , 10/15	Self-supervised Leaning	AutoEncoder (AE), Variational AE, Masked AE, Contrastive learn-ing, CLIP, DINO
		W9 10/20, 10/22	Foundation Models	Large Language Models (LLMs), Vision Language Models (VLMs)
		W10 10/27, 10/29	Generative Models	Text-to-image and text-to-video generation with diffusion models
Thanksgiving Holiday	Module 3: CV for Engineering Systems	W11 11/3, 11/5	Geometry & 3D Vision	3D Representations, Modern 3D Reconstruction & Generation Models (NeRF, 3DGS, VGGT, TRELLIS, Marble); Spatial Intellignece
		W12 11/10, 11/12	Robotics & Autonomous	Robotic perception loops; VLA, navigation, embodied reasoning, motion planning & diffusion policy
		W13 11/17, 11/19	Remote Sensing & Earth Observation	Satellite imagery; multispectral/hyperspectral; geospatial segmenta-tion; continuous temporal monitoring
		W14 11/24, 11/26	Mining & Energy	Industrial anomaly detection: wind turbine cracks, PV panel, gas / oil / power lines
		W15 12/1, 12/3	Climate & Environment	Disaster perception (flood, landslide, glacier, wildfire); Weather per-ception; Urban scene reconstruction
		W16 12/8	Healthcare	Medical image analysis & diagnosis (X-ray, MRI, pathology)
	Final Exam	W17 12/02?	Project Presentation & Poster Exhibition	Project Roadshow: 12 students (4 or 6 groups) presenting application-focused projects built on real-world datasets / their research

Lectures

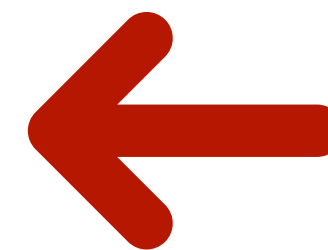
- **Tuesdays and Thursdays** between **12.30pm to 1:45pm** at **Brown Building W475**
- **Slides and Code** will be posted on *Canvas & Github repo* shortly before each lecture:
 - <https://github.com/MinghanLi/CSCI598H-Applied-Computer-Vision-for-Engineering-Systems>
- All additional materials and resources on *Canvas*
 - Syllabus, course materials, GPU guide, assignments, Grades and etc.

Friday Discussion/Code Sections (optional)

- Check the course website for time and location

Sept/04	Access to GPU / Python / Numpy
Sept/18	CNNs / Transformers
Oct/09	Segmentation / LLMs
Oct/30	Diffusion Model / NeRF
Nov/20	Engineering Applications

- Hands-on tutorials, with more practical details than the main lecture



**Scan QR code
to add your
available time!**

<https://www.when2meet.com/?38164088-qI3Qc>

Friday Discussion/Code Sections (optional)

- Check the course website for time and location

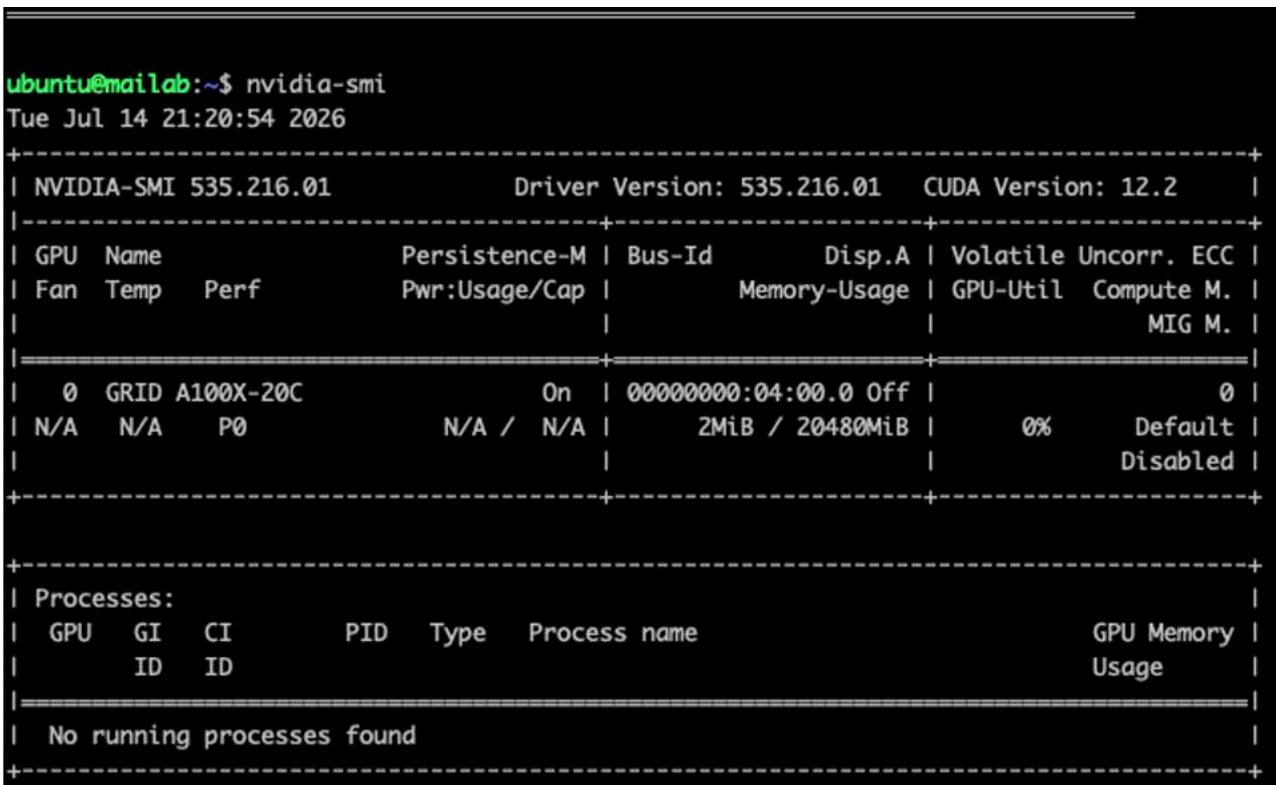
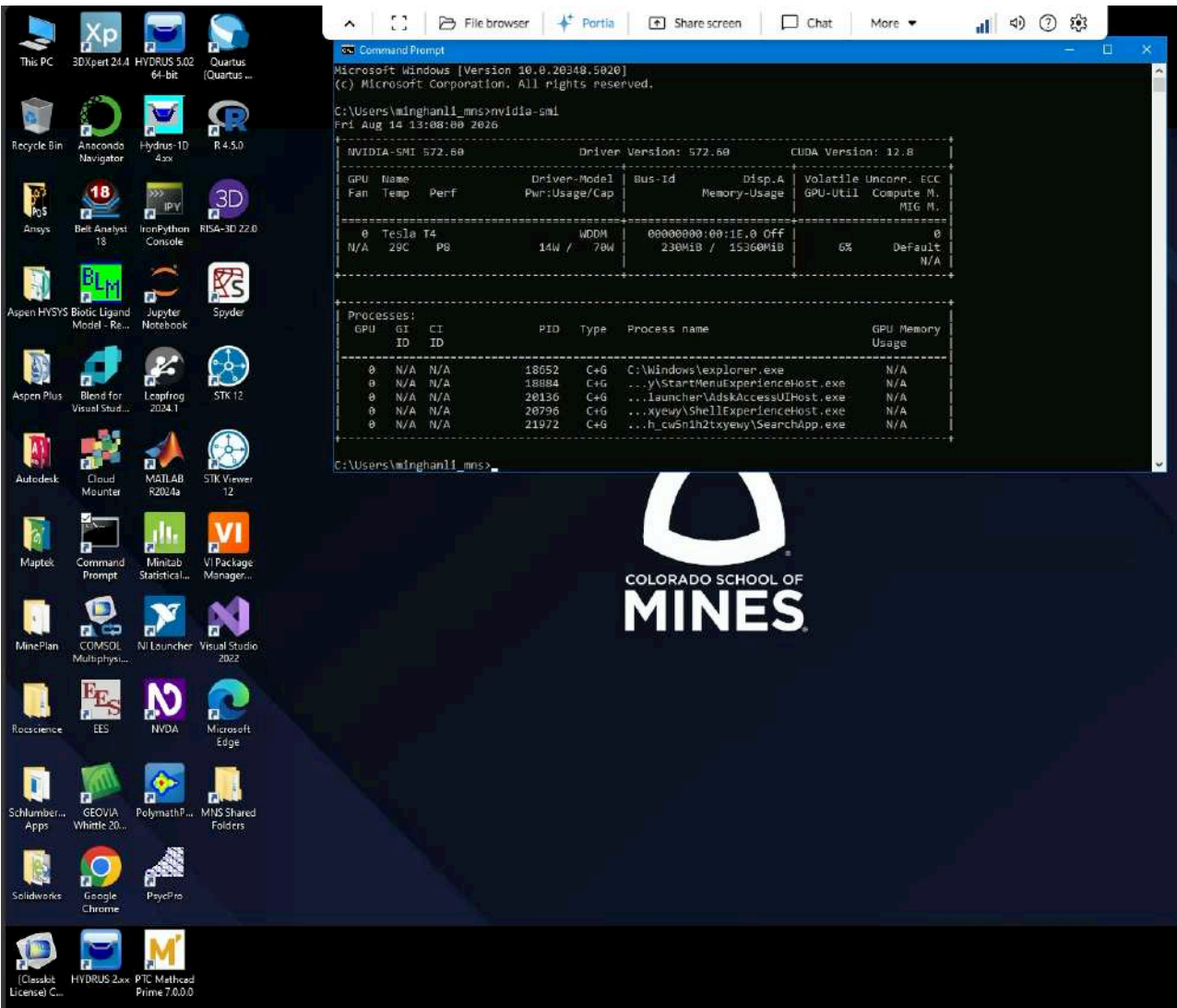
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- Hands-on tutorials, with more practical details than the main lecture

- GPU support:

- Mines GPU Desktop: Canvas → CSCI598H → GPU Desktop
- NSF ACCESS (JetStream2): [User Guide](#)

- Next Friday: Access to GPU / Python / PyTorch / Numpy



Question and Office Hours

- We'll be hosting both in-person and remote office hours
 - Location & Time
 - In-person: CTLM 214A, Tuesday & Thursdays between 2pm to 4pm
 - Remote: send emails to minghan.li@mines.edu
- Use private questions only if your post will violate the honor code if you release it publicly

Assignments, Exam, Course Project, Grading

Attendance: 15%

3 Assignments: 10% + 10% + 15% = 35%

In-Class Paper Reading: 20% (W12-W16 / highly related to your research)

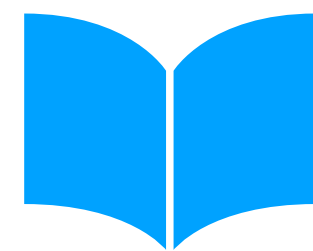
Course Final Project: 30% (based on paper reading → challenges → your attempts)

- Project Proposal: 2%
- Milestone Check-Ins: 2% each = 6%
- Final Project Report & Code: 17%
- Poster Exhibition: 5% (will invite students from CS, Robotics, Engineering, ...)



~W10

(YOU share me the
paper title)



W12-W16

(Paper reading)



W16

(Project proposal
+ poster)

Try to

- identify the main challenges
- Propose your ideas
- Implement & get some initial results

Assignments, Exam, Course Project, Grading

Attendance: 15%

3 Assignments: 10% + 10% + 15% = 35%

In-Class Paper Reading: 20% (W12-W16 / highly related to your research)

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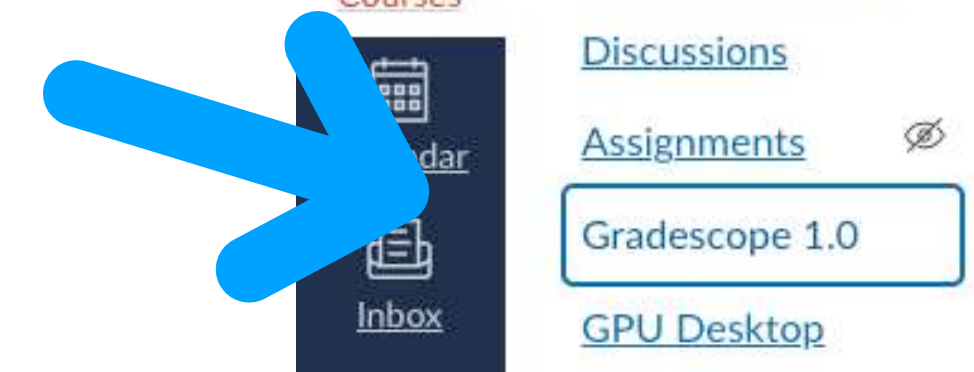
Participation Extra Credit: up to 3%

Late Policy

- 4 free late days — use up to 2 late days per assignment
- Afterward, 25% off per day late
- No late days for project report

Assignments, Exam, Course Project, Grading

All assignments, coding, and written portions will be submitted via *Gradescope 1.0*



- All assignments have a **written and a coding section**
- The code section is intended to get students' hands-on experience with the gritty implementation details of widely used algorithms in AI/CV. The submission will be **autograded**.
- The written submission is intended to build intuition and conceptual understanding about the algorithms. The submission will be **graded by Dr. Minghan Li**.
- **AI Tools Policy. Students may use AI tools (e.g., ChatGPT, Claude, GitHub Copilot) only for limited learning support, such as clarifying concepts or debugging general errors. The following uses are prohibited:**
 - **Generating or completing assignment code**
 - **Submitting AI-generated text, analysis, figures, or results**
 - **Using AI to substantially rewrite another person's work**

All submitted work must be the student's own. Unauthorized use will be treated as an academic-integrity violation.

Collaboration Policy

We follow the *Mines Student Honor Code and Academic Integrity/Misconduct* —read them carefully!

- **Rule 1:** Don't look at solutions or code that are not your own (including AI tools); everything you submit should be your own work.
- **Rule 2:** Don't share your solution code with others; however, discussing ideas or general strategies is fine and encouraged.
- **Rule 3:** Indicate in your submissions anyone you worked with.
- **Rule 4:** Do not submit AI-generated responses.

Turning in something late or incomplete is better than violating the honor code.

Assignments 35%

Proficiency in Python

- All class assignments will be in Python (and use NumPy).
- Later in the class, you will use PyTorch.
- A Python tutorial is available here ([Stanford tutorial](#)). Also, we will have code sections before Assignment 1.

Assignment 1 (10%): will be out *Sept/17/2026 12AM on Gradescope 1.0*

- Linear classifier: Softmax
- Two-layer neural network
- Image features
- Deep neural network and optimizers
- TBD (*not the final version*)

Optional Textbook Resources

Deep Learning

- By Goodfellow, Bengio, and Courville
- Free version available online

Mathematics of Deep Learning

- Chapters 5–7 are useful for understanding vector calculus and continuous optimization
- Free online version



Dive into Deep Learning

- An *interactive deep learning book* with code, mathematics, and discussions based on the NumPy interface
- Free online version

Learning Objectives

- **Formalize real-world computer vision problems**
 - Define appropriate inputs, outputs, tasks, datasets, and evaluation metrics.
 - Identify the data and computational resources required to develop effective solutions.
- **Develop and train deep vision models**
 - Implement, debug, train, and evaluate neural networks using Python and PyTorch.
 - Apply methods and architectures for classification, detection, segmentation, and other vision tasks.
- **Understand modern computer vision paradigms**
 - Explain recent advances in generative models, self-supervised foundation models, and 3D neural rendering.
 - Critically analyze model capabilities, limitations, and open research challenges.
- **Apply computer vision to engineering and science**
 - Develop vision solutions for robotics, remote sensing, climate monitoring, mining, infrastructure inspection, and structural bio-AI.
 - Collaboratively design, evaluate, and present a research-focused final project.

Why should you take this class?

Get involved with vision research (An incomplete list of conferences and journals)

- MAILab: apply using this form
- CVPR 2027 Conference
- ICCV 2027 Conference

| Featured post

Upcoming CVF Sponsored Conferences

admin June 3, 2026

[WACV 2027](#): January 4th – 8, Orlando, Florida

CVPR 2027: June 20th – 24th, Seattle, Washington

ICCV 2027: October 2nd – 8th, Hong Kong

Become a vision engineer in industry (An incomplete list of industry teams)

- Perception Team at Google AI
- Vision at Meta AI
- Vision at Amazon AWS
- NVIDIA, Apple, Microsoft, OpenAI, Salesforce, Alibaba, TikTok, ...

Apply computer vision to solve problems in engineering systems (next page)

Why should you take this class?

Computer Vision and AI Conferences

- CVPR, ICCV, ECCV
- NeurIPS, ICML, ICLR
- WACV, AAAI, ACM Multimedia

Engineering and Applied-Vision Conferences

- ICRA, IROS, RSS — Robotics
- IGARSS — Remote Sensing
- IEEE ICIP — Image Processing
- MICCAI — Medical Imaging
- IEEE CASE — Automation and Engineering Systems

Major Journals

- IEEE TPAMI, IJCV — Computer Vision
- IEEE TIP, IEEE TMM — Image Processing and Multimedia
- IEEE TGRS, ISPRS Journal — Remote Sensing
- IEEE T-RO, RA-L — Robotics
- IEEE TMI, Medical Image Analysis — Medical Imaging

Course Feedback & Continuous Improvement

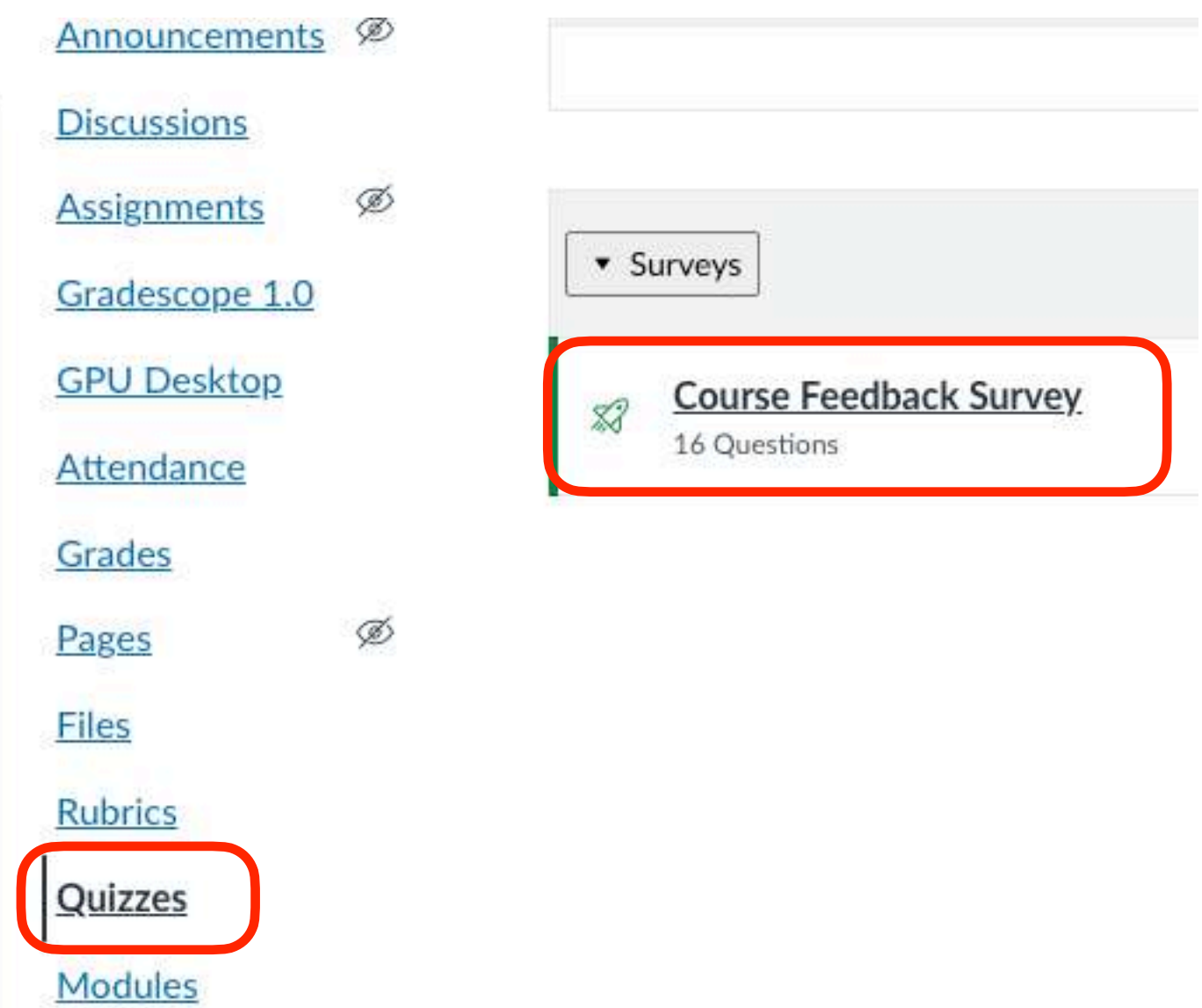
Your feedback helps shape this course.

Why?

- Understand whether the **pace and difficulty** are appropriate
- Identify topics that need **more explanation or practice**
- Improve the balance between **theory, CV methods, and engineering applications**
- Evaluate the usefulness of **lectures, examples, assignments, and projects**
- Make adjustments **during the semester**, not only after the course ends

How?

- Canvas → CSCI 598H → Quizzes → Course Feedback Survey

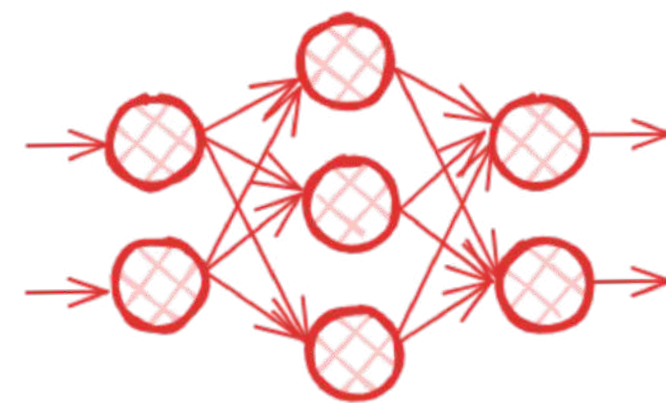


Next Lecture: Deep Learning Basics

Image Classification with a linear classifier



Images



0.5 Airplane

0.2 Bird

0.3 Cat



**Loss
Function**

0

1

0

Labels

Predictions

Backpropagation

Linear classification

